Financial Models

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The Value Paradox

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Nvidia versus Bitcoin

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Best Strategy to Develop of Flu Vaccine

Column1	Column2	Column3	Column4	Column5
Time	**0**	**1**	**2**	**3**
Strategy A:	-$1,000	$500	$400	$300
Strategy B:	-$200	-$200	$300	$300

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What is Financial Model?

- A financial model is simply **a decision-making tool (e.g., buy/sell stocks; invest/abandon/expand/contract projects)** that is built in Excel to project an entity's financial performance into the future.

- The modelling process is basically an integration of technical (Excel), academic (Accounting Finance, Statistics, Engineering), and business skills to create a user-friendly decision-making tools.

- A good financial model is a powerful communication tool to make more informed and effective financial decisions.

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Building A Financial Model

- Keep it simple!
- It is highly critical to have strong plan before you start building your model.
  - What is the purpose of the model?
  - What are the critical assumptions underlying your financial model?
  - What are the critical inputs in your financial model?
  - How would you forecast the future evolution of financial variables in the model? How many periods should be included in the forecast?
  - What are the main outputs that your client would be interested in?
- Gather relevant information to feed into your model.
- Remember, keep it simple. A financial model is a means to an end, not the end itself.

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Types of Financial Models

- Most financial models focus on valuation.
- Some models are created to calculate and predict credit risk, performance of investment portfolio, or economic trends within an industry or a country.
- **DCF**: Among different types of Financial model, DCF Model is the most important one. It is based upon the theory that the value of a business is the sum of its expected future free cash flows discounted at an appropriate rate.
- **COMPARABLE**: Also referred to as "Relative Valuation", it is the one of the major company valuation analyses that is used in the investment banking industry. In this method we undertake a peer group analysis under which we compare the financial metrics of a company against similar firms in industry. It is based on an assumption that similar companies would have similar valuations multiples, such as EV/EBITDA.

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Types of Financial Models

- **SUM-OF-THE-PARTS:** It is also referred to as the break-up analysis. This modeling involves valuation of a company by determining the value of its divisions if they were broken down and spun off or they were acquired by another company.
- **LEVERAGED BUY OUT (LBO) :** The LBOs typically involves acquiring another company using a significant amount of borrowed funds. This kind of model is being used mostly in leveraged finance at Private Equity firms who want to acquire companies with an objective of selling them in the future for a hefty profit.
- **MERGERS & ACQUISITION (M&A):** The objective of M&A modeling is to show clients the impact of an M&A to the acquirer's EPS. When the new EPS (after M&A) is higher, the transaction is called "accretive" while the opposite is called "dilutive."
- **CONTINGENT CLAIM (OPTION):** Uses option pricing models to measure the value of assets that share option characteristics.

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Probabilistic Approaches in Financial Modelling

- All financial models can be classified into either of the two categories:

- **Static Financial Models:**
  - All model parameters are given and the outcome from an investment deterministic.
  - Example: Constant growth DDM.

- **Dynamic Financial Models:**
  - Model parameters are changing but are known well in advance.
  - Example: Multi-stage DDM or Residual Income Valuation model.

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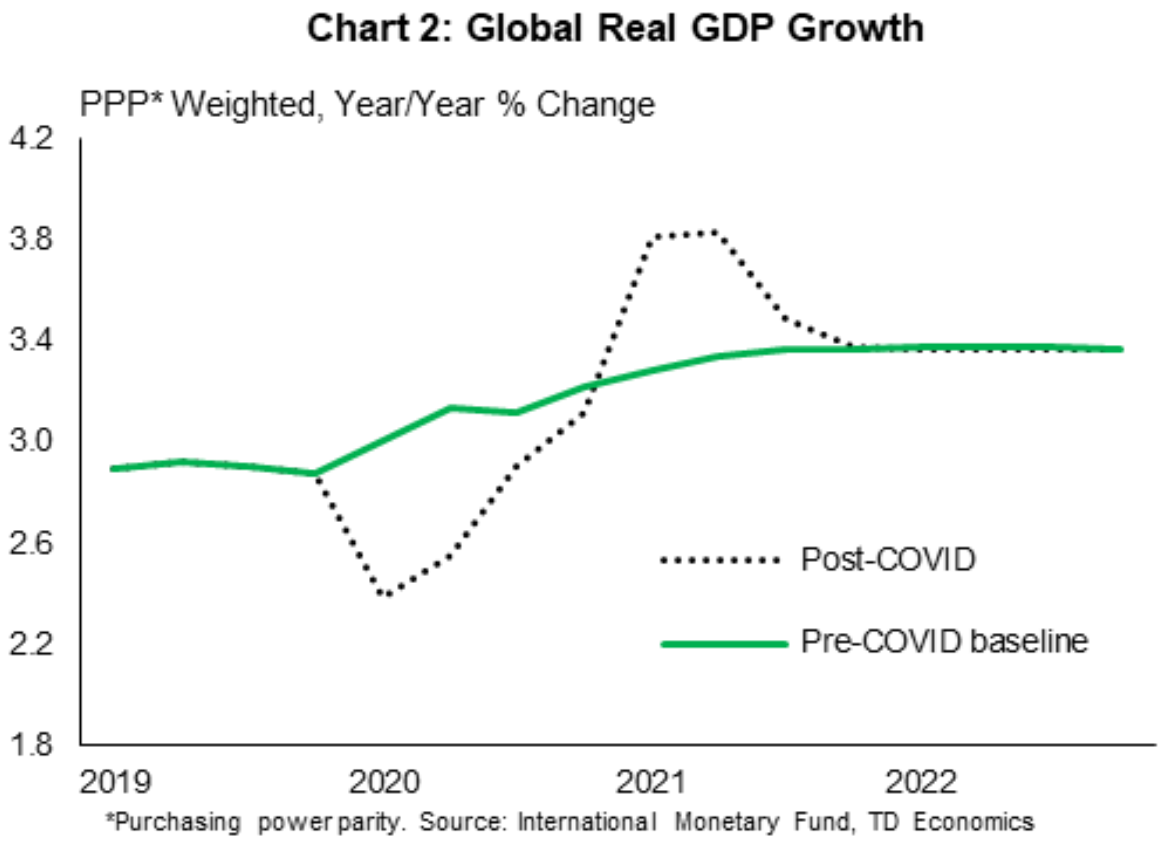
Financial Model VS Reality



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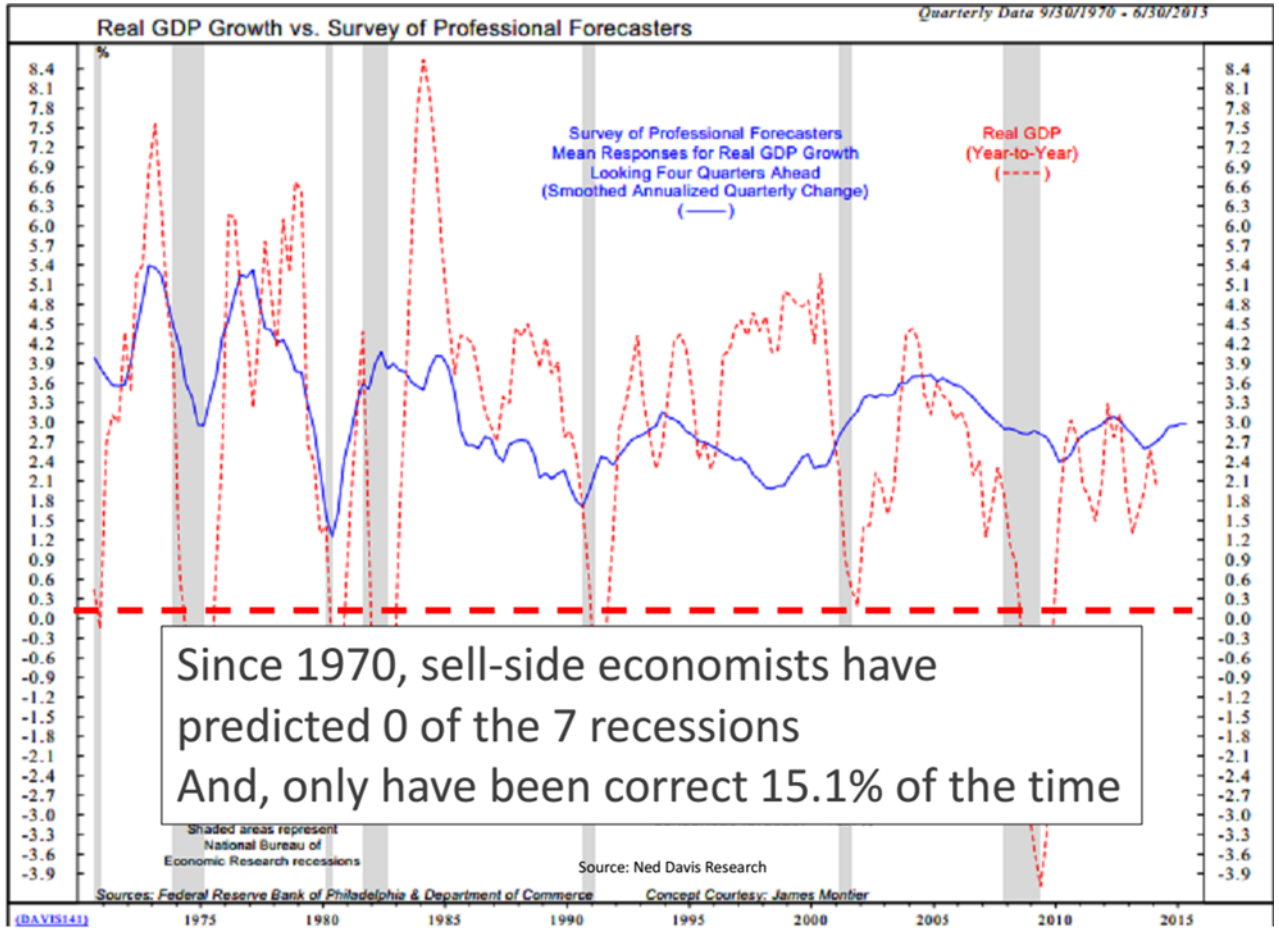
Financial Model vs Reality



Since 1970, sell-side economists have predicted 0 of the 7 recessions
And, only have been correct 15.1% of the time

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Building Blocks of Financial Models: The Time Value of Money

Value of Money in Finance

- Tax Value
  - How much of your money goes to the government in paying taxes?
  - Can you increase your firm value by minimizing your tax burden?
- Risk Value
  - How much risk you had to take to get the money?
  - Did you receive fair compensation for the risk you have under taken?
- Time Value
  - When did you get your money?
  - Does it matter to you whether you get your salary today or at the end of the year?
- Value of Money=Tax Value + Risk Value + Time Value

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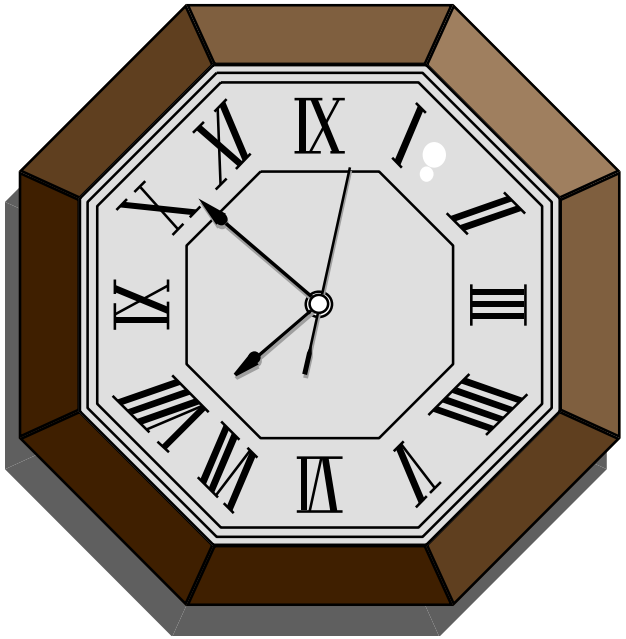
Why Time?

- Time is Money.

- If you $10000 student loan and you take 10 years to repay the loan at 7.5% interest, you will have paid $14200 (142%) of the value of the loan.

- "The greatest mathematical discovery of all time is compound interest."

  Albert Einstein



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What is Time Value?

- A Big Mac was $0.49 in 1968 when first introduced, in 2023 the same Big Mac is $529.
- A bottle of Coke was $0.10 in 1950, $0.59 in 1995, and $2.99 in 2022.
- **Why?**

- A dollar today is not necessarily what a dollar is worth tomorrow.
  - Would you trade $1 today for $1 ten years from now?
- Giving up money has an **opportunity cost**.
- Opportunity cost = Alternative use
  - The opportunity cost of money is the interest rate that would be earned by investing it.
  - This is the underlying reason for the time value of money.
  - Money today can be invested to be some greater amount in the future.
  - Conversely, if you are promised a cash flow in the future, it's present value today is less than what is promised!

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What is Time Value?

- Time value of money involves two components:
  - Cash flow:
    - What is the nature of the cash flow?
    - Is it real or nominal?
    - Is it a single cash-flow or a stream of cash flows?
    - Is it a normal cash-flow stream or infrequent cash-flow stream?
    - What is frequency of cash flows, i.e., yearly, bi-yearly, monthly, weekly, daily, continuously?
  - Interest rate:
    - Is it real or nominal rate?
    - What is the compounding frequency of the rate?
    - It is an annual percentage rate (APR) or effective annual rate?

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Time Value and Interest Rates

- The Interest rate (also known as discount rate) is used to convert cash across time.
  - Interest Rate/discount rate (r)
    - The rate at which money can be borrowed or lent over a given period
  - Interest rate/discount rate factor (1 + r)
    - It is the rate of exchange between dollars today and dollars in the future.

- The appropriate interest rate depends on:
  - the pure rate of interest (Prime Rate)
  - credit risk rate of interest
  - expected inflation rate of interest

- The higher the prime rate, the higher the interest rate.
- The higher the credit risk, the higher the interest rate.
- The higher the inflation, the higher the interest rate.

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Interest Computation

- Principal: The amount borrowed or invested
- Interest rate: A percentage of the outstanding principle.
- Time: the number of years or fractional portion of a year that principal is outstanding.

  - Simple interest is determined on the principal only.
    - Principal x interest rate (%) x time
  - Compound interest is determined on: the principal, and any interest earned (and not withdrawn). In essence, interest earns interest.

- Compound interest is the typical computation applied in most time value applications.

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Interest Rates and Compounding Frequency

Assumed interest rate per year: 12%

Frequency of Compounding	Interest rate per compounding	Number of compounding periods
Annual	12%	One (1)
Semi-annual	6%	Two (2)
Quarterly	3%	Four (4)
Monthly	1%	Twelve (12)

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Time Value and Interest Rates

- Value of $100,000 Investment in One Year
  - If the interest is 10%, the value of the investment in one year is:
  - Value = ($100,000 today) × (1.10 $ in one year/$ today)

    = $110,000 in one year

- The Interest rate/discount rate: Converting cash across time
  - $110,000 is the opportunity cost of spending $100,000 today
    - The firm gives up the $110,000 it would have had in one year if it does not invest the money today.
    - Alternatively, by borrowing the $100,000 from the bank, the firm would owe $110,000 in one year. Because by lending you $100,000 today the bank gives up other investment opportunity, so you need to compensate the bank for that by repaying not $100,000 but $110,000.

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Time Value and Interest Rates

- The Interest rate/discount rate: Converting cash across time
  - Present Value: The value of a cost or benefit computed in terms of cash today
  - Future Value: The value of a cash flow that is moved forward in time

- Two technologies to convert cash across time
  - Discount Factors:
    - Money in the future is worth less today so its price reflects a discount.
    - Discount Rate: The appropriate rate to discount a cash flow to determine its value at an earlier time: r
    - Discount Factor: The value today of a dollar received in the future, expressed as:

$$\boxed{\frac{1}{1+r}}$$

  - Compound Factors:
    - Money saved today worth more in the future.
    - Compounding Factor: The value in the future of a dollar saved today, expressed as:

$$(1+r)$$

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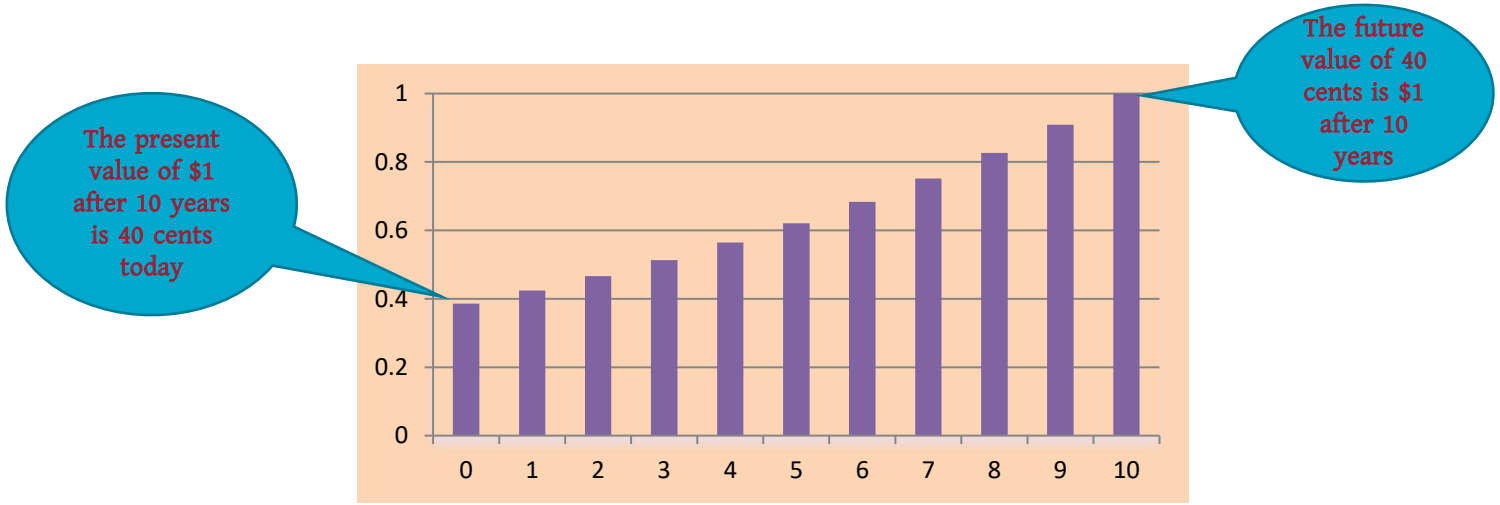
Time Value: Discounting/Compounding

- **Discounting** is the process of turning future cash flows into current cash flows. It is also known as **Present Value**.
  - Example: A dollar ten years from now is worth less than 40 cents today if the market interest rate is 10%.

- **Compounding** is the process of turning current cash flows into future cash flows. It is also known as **Future Value**.
  - Example: 40 cents today is worth a dollar ten years from now if the market interest rate is 10%.



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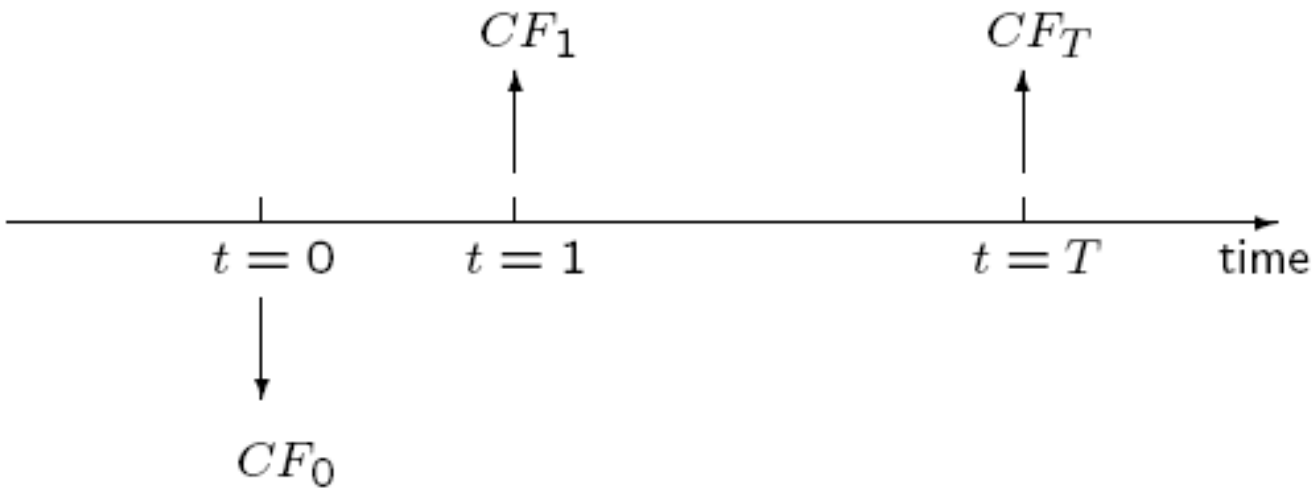
Valuing Cash-Flow Over Time

"Visualizing" cash flows.



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Valuing Cash-Flow Over Time

- Example: Drug company develops a COVID-19 vaccine.
  - Strategy A: To bring to market in 1 year, invest $1B (billion) now and returns $500M (million), $400M and $300M in years 1, 2 and 3 respectively.
  - Strategy B: To bring to market in 2 years, invest $200M in years 0 and 1. Returns $300M in years 2 and 3.

- Which strategy creates more value?

- Problem: How to compare two Cash-flow (CF) streams?
- Solution: For each CF stream, find a single number:
  - Equal in value to the cash-flow stream.
  - Valued on the same date (e.g. today) for all cash flows.
  - Use this number to compare different CF streams.

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Solution to Example: Flu Vaccine

Assume that $r = 5\%$.

Strategy A:

Time	0	1	2	3
Cash Flow	-1,000	500.0	400.0	300.0
Present Value	-1,000	476.2	362.8	259.2
			Total PV	98.2

Strategy B:

Time	0	1	2	3
Cash Flow	-200	-200.0	300.0	300.0
Present Value	-200	-190.5	272.1	259.2
			Total PV	140.8

Firm should choose strategy B, and its value would increase by \$140.8 M.

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Future Value

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Future Value: One-Period Case

- If you were to invest $10,000 at 5-percent interest for one year, your investment would grow to $10,500
  - $ 500 would be interest ($10,000 × .05)
  - $10,000 is the principal repayment ($10,000 × 1)
  - $10,500 is the total due. It can be calculated as:
  - $10,500 = $10,000×(1.05).

- The total amount due at the end of the investment is called the **Future Value (FV).**

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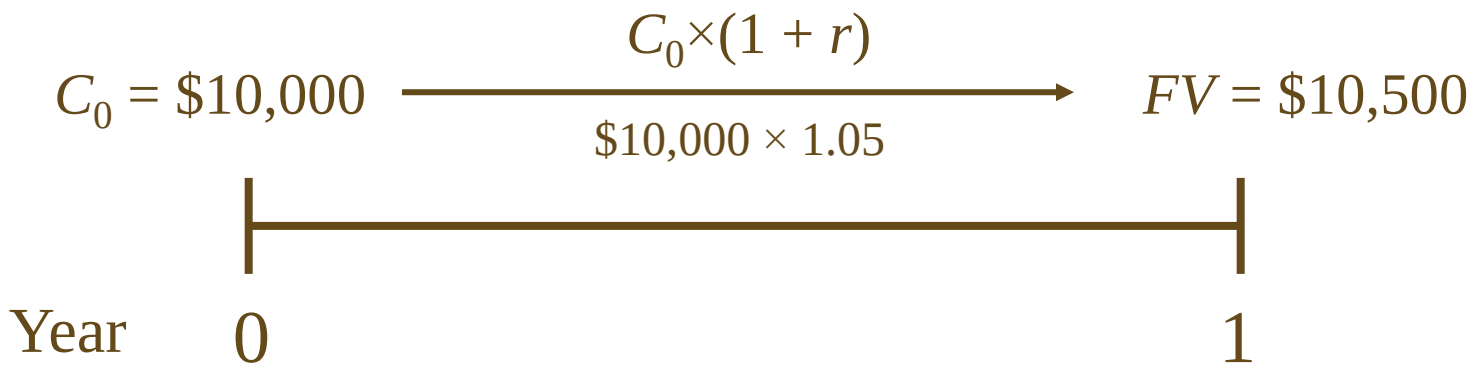
Future Value: One-Period Case

- In the one-period case, the formula for FV can be written as:

$$FV = C_0 \times (1 + r)$$

  - Where C_0 is cash flow at date 0 and r is the appropriate interest rate (discount rate, opportunity cost of capital, required rate of return, cost of capital, hurdle rate).



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Future Value: Multi-Period Case

- The general formula for the future value of an investment over many periods can be written as:

$$FV = C_0 \times (1+r)^T$$

  - C_0 is cash flow at date 0,
  - r is the appropriate interest rate, and
  - T is the number of periods over which the cash is invested.

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Future Value: Multi-Period Case

- You invest $100 at 40% per year for 5 years
  - Interest is paid at the end of each year
  - When interest is paid, re-invest it at 40%
- How much will the investment be worth at the end of each year?

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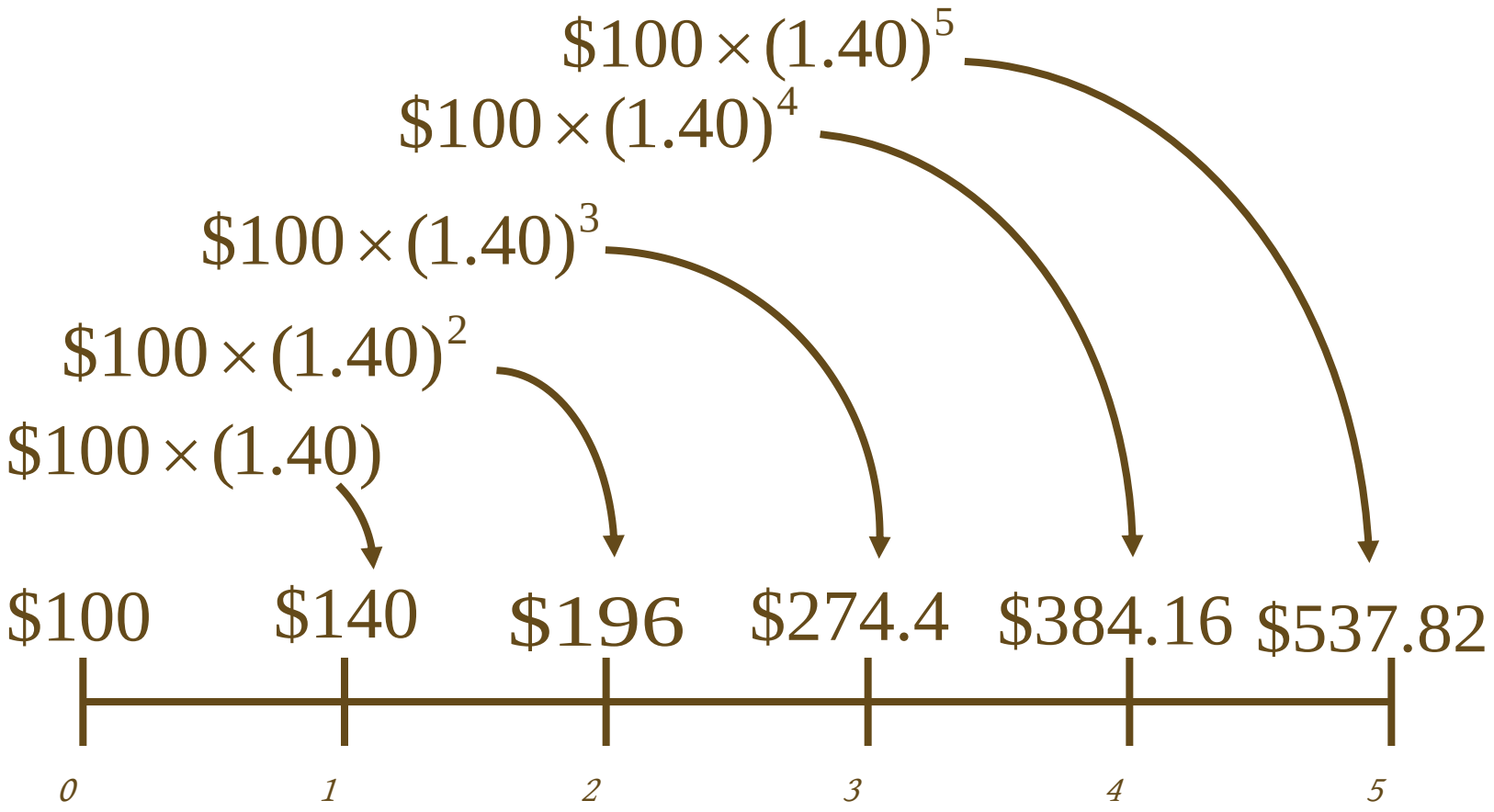
Future Value and Compounding



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Future Value and The Power of Compounding

- Notice that the value of the investment in year five, \$537.82, is considerably higher than the sum of the original investment plus five interest payments of 40-percent on the original \$100 investment:
  - $\$537.82 > \$300 = \$100 + 5\times[\$100\times0.40]$

- This is due to **compounding.**
  - The interest at the end of each year is reinvested and earns its own interest.
  - With no compounding (i.e. no reinvestment of interest) the total interest on the investment is: $5\times (100\times0.4) = 200$.

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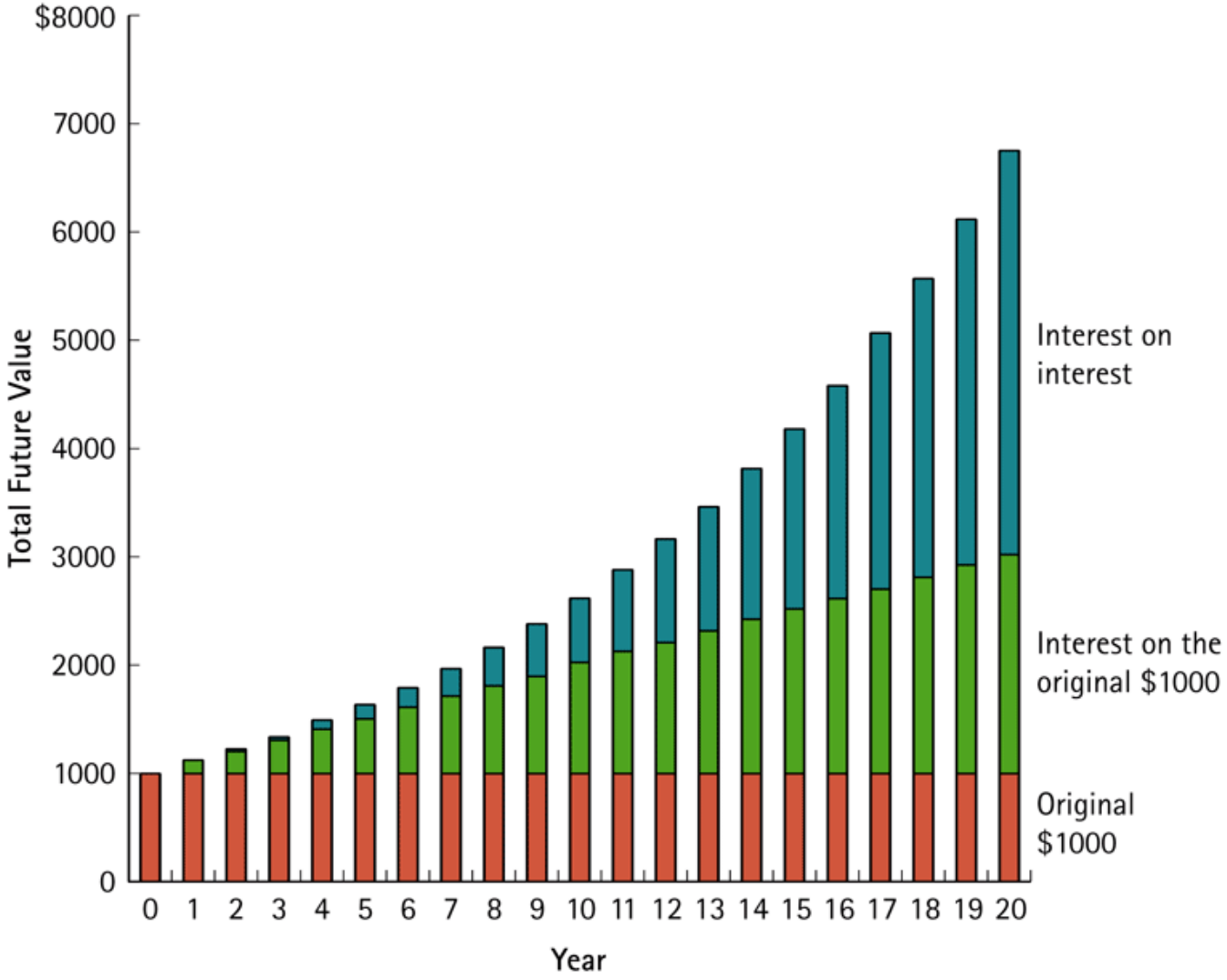
The Magic of Compounding



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Compounding Frequency

- Compounding an investment m times a year for T years provides for future value of wealth:

$$FV = C_0 \times \left(1 + \frac{r}{m}\right)^{m \times T}$$

- For example, if you invest \$50 for 3 years at 12%
  - compounded annually, your investment will grow to:
    - FV = \$50 × $(1 + .12/1)^{1\times3}$ = 50 × $(1.12)^3$ = \$ 70.25
  - Compounded semi-annually:
    - FV = \$50 × $(1 + .12/2)^{2\times3}$ = 50 x $(1.06)^6$ = \$ 70.93
  - Compounded monthly:
    - FV = \$50 × $(1 + .12/12)^{12\times3}$ = 50 x $(1.01)^{36}$ = \$ 71.54

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Effective Annual Interest Rates

- There is a difference between the Stated vs. Effective Interest Rates. Although the convention is to quote interest rates on annual basis but may be compounded more frequently: semi-annual, quarterly, daily, etc.

- The reasonable question to ask in the above example is what is the effective annual rate of interest on that investment if the investment is compounded semi-annually.

- The Effective Annual Interest Rate (EAR) is the annual rate that would give us the same end-of-investment wealth after 3 years:

$$FV = \$50 \times (1 + \frac{.12}{2})^{2\times 3} = \$50 \times (1.06)^6 = \$70.93$$

$$\$50 \times (1 + EAR)^3 = \$70.93$$

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Effective Annual Interest Rates

$$FV = \$50 \times (1 + EAR)^3 = \$70.93 \Rightarrow (1 + EAR)^3 = \frac{\$70.93}{\$50}$$

$$EAR = \left(\frac{\$70.93}{\$50}\right)^{1/3} - 1 = .1236 \text{ or } 12.36\%$$

- So, investing at 12.36% compounded annually is the same as investing at 12% compounded semiannually.

$$EAR = \left(1 + \frac{r_s}{m}\right)^m - 1$$

- EAR = Effective Annual Rate
- r_s= Stated annual interest rate or APR

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Continuous Compounding

- The general formula for the future value of an investment (C_0) compounded continuously (i.e. as $m \rightarrow \infty$) over many periods can be written as:

$$FV = C_0 \times e^{r_s T}$$

  - C_0 is cash flow at date 0, C_T is cash flow at date T,
  - r_s is the stated annual interest rate,
  - T is the number of periods over which the cash is invested, and
  - e is a transcendental number approximately equal to 2.718.

- The effective annual interest rate

$$r_e = e^{r_s} - 1$$

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Continuous Compounding: An Example

- For example, if you invest \$50 for 3 years at 12% annual percentage rate (APR) your investment will grow to:

- Compounded annually:
  - $FV = \$50 \times (1 + .12/1)^{1\times3} = 50 \times (1.12)^3 = \$\ 70.25$
- Compounded semi-annually:
  - $FV = \$50 \times (1 + .12/2)^{2\times3} = 50 \text{ x } (1.06)^6 = \$\ 70.93$
- Compounded monthly:
  - $FV = \$50 \times (1 + .12/12)^{12\times3} = 50 \text{ x } (1.01)^{36} = \$\ 71.54$
- Compounded continuously:
  - $FV = \$50 \times e^{0.12\times3} = 50 \text{ x } e^{0.36} = \$\ 71.67$

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Present Value

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Present Value: One-Period Case

- If you were to be promised $10,000 due in one year when interest rates are at 5-percent, your investment would be worth $9,523.81 in today's dollars.

$$\$9{,}523.81 = \frac{\$10{,}000}{1.05}$$

- The amount that a borrower would need to set aside today to be able to meet the promised payment of $10,000 in one year is call the **Present Value (PV)** of $10,000.

Note that $10,000 = $9,523.81×(1.05).

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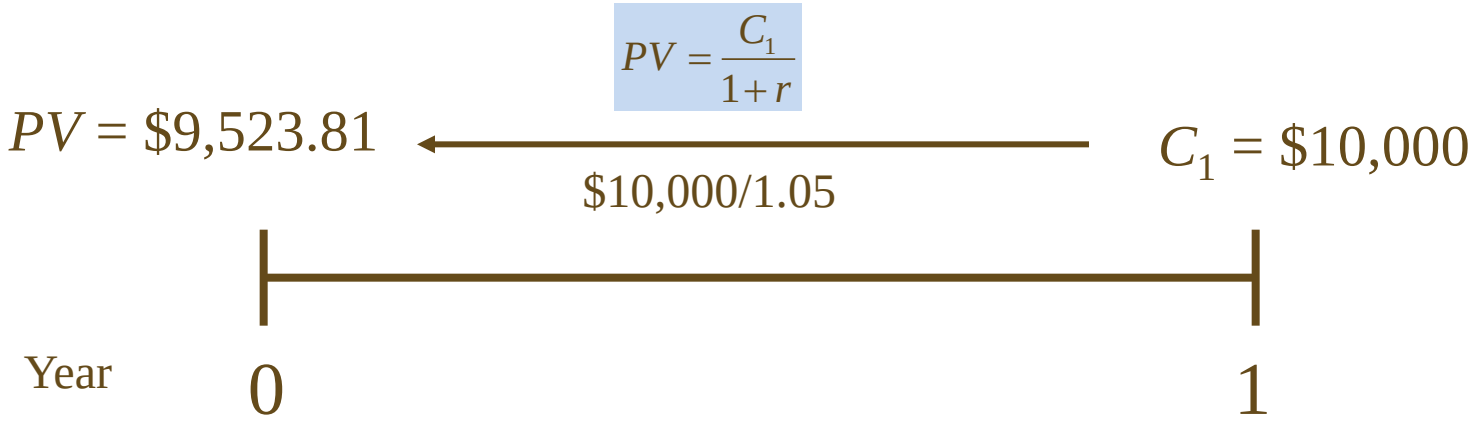
Present Value: One-Period Case

- In the one-period case, the formula for PV can be written as:

$$PV = \frac{C_1}{1+r}$$

Where C_1 is cash flow at date 1 and r is the appropriate interest rate.



Net Present Value: One-Period Case

- The Net Present Value (*NPV*) of an investment is the present value of the expected cash flows, less the cost of the investment.
- Suppose an investment that promises to pay \$10,000 in one year is offered for sale for \$9,500. Your interest rate is 5%. Should you buy?

$$NPV = -\$9{,}500 + \frac{\$10{,}000}{1.05}$$
$$NPV = -\$9{,}500 + \$9{,}523.81$$
$$NPV = \$23.81$$

Yes!

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Present Value: Multi-Period Case

- The general formula for the future value of an investment over many periods can be written as:

$$PV = \frac{C_T}{(1+r)^T}$$

  - C_0 is cash flow at date T,
  - r is the appropriate interest rate, and
  - T is the number of periods.

- How much would an investor have to set aside today in order to have $20,000 five years from now if the current rate is 15%?

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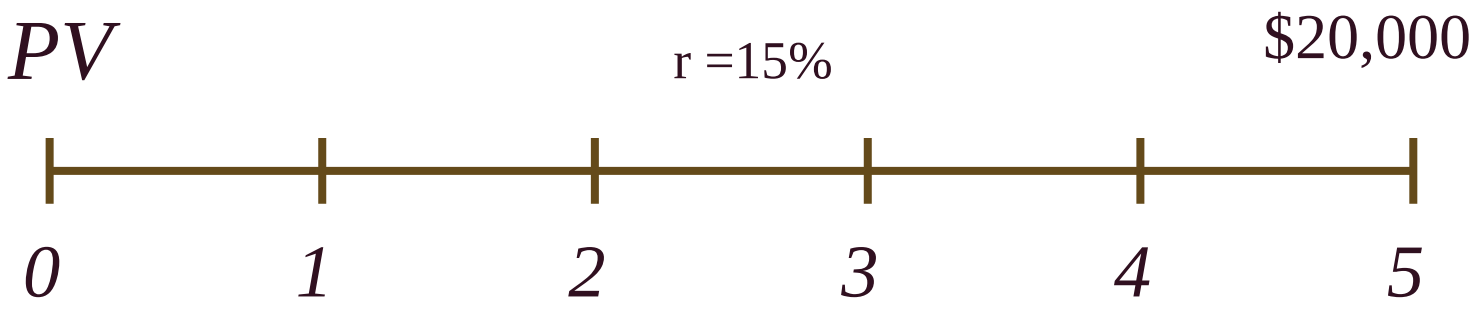
Present Value and Discounting



$$PV = \$9{,}943.53 = \frac{\$20{,}000}{(1.15)^5}$$

What if r = 10%?

$$PV = \$12{,}418.43 = \frac{\$20{,}000}{(1.1)^5}$$

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Continuous Discounting

- The general formula for the present value of a future payment (C_T) with continuous discounting (i.e. as $m \rightarrow \infty$) over many periods can be written as:
  - $PV = C_T \times e^{-rT} = C_T / e^{rT}$
  - C_0 is cash flow at date 0, C_T is cash flow at date T,
  - r is the stated annual interest rate,
  - T is the number of periods over which the cash is invested, and
  - e is a transcendental number approximately equal to 2.718.

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How Long is the Wait?

- If we deposit \$5,000 today in an account paying 10%, how long does it take to grow to \$10,000?

$$FV = C_0 \times (1 + r)^T$$

$$\$10{,}000 = \$5{,}000 \times (1 + 0.10)^T$$

$$(1.10)^T = \frac{\$10{,}000}{\$5{,}000} = 2$$

$$T \times \mathrm{Log}(1.10) = \mathrm{Log}(2)$$

$$T = \frac{\mathrm{Log}(2)}{\mathrm{Log}(1.10)} = 7.27$$

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What Rate Is Enough?

- The total cost of a university education will be \$50,000 when your child enters university in 12 years. You have \$5,000 to invest today. What rate of interest must you earn on your investment to cover the cost of your child's education?

$$FV = C_0 \times (1 + r)^T$$

$$\$50{,}000 = \$5{,}000 \times (1 + \mathrm{r})^{12} \Rightarrow (1 + \mathrm{r})^{12} = \frac{\$50{,}000}{\$5{,}000} = 10$$

$$\Rightarrow \mathrm{r} = (10)^{\frac{1}{12}} - 1 = .2115 = 21.15\%$$

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Valuing Cash Flow Stream

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Valuing a Stream of Cash Flows

- Rules:
  - Rule 1: Only values at the same point in time can be compared or combined. That means, you can compare present value with another present value. You cannot compare present value with future value.
  - Rule 2: To calculate a cash flow's future value, we must compound it.
  - Rule 3: To calculate the present value of a future cash flow, we must discount it.

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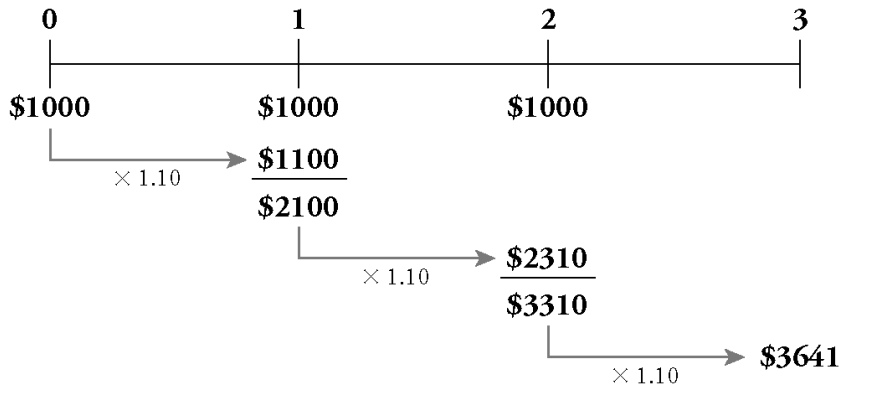
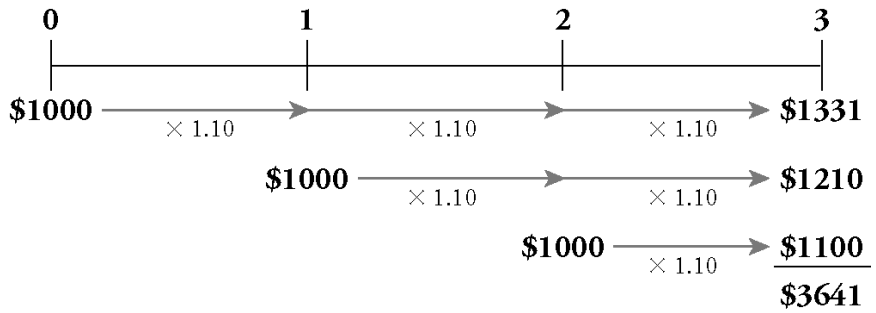
Valuing a Stream of Cash Flows

- Suppose you plan to save $1,000 today, and $1,000 at the end of each of the next two years. If we earn a fixed 10% interest rate on our savings, how much will we have three years from today?





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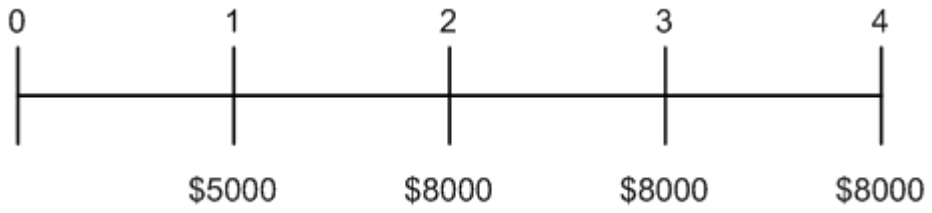
Valuing a Stream of Cash Flows

- What is the present value of the following cash flows stream if the current market interest rate is 6%?



$$
\begin{aligned}
PV &= \frac{5000}{1.06} + \frac{8000}{1.06^2} + \frac{8000}{1.06^3} + \frac{8000}{1.06^4} \\
&= 4716.98 + 7119.97 + 6716.95 + 6336.75 \\
&= \$24,890.65
\end{aligned}
$$

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Special Cash Flow Streams

- Perpetuity
  - A constant stream of cash flows that lasts forever
- Growing (delayed) perpetuity
  - A stream of cash flows that grows at a constant rate forever
- Annuity
  - A stream of constant cash flows that lasts for a fixed number of periods
- Growing annuity
  - A stream of cash flows that grows at a constant rate for a fixed number of periods
- Infrequent annuity
  - A stream of cash flows for a fixed number of periods but come in an infrequent fashion.

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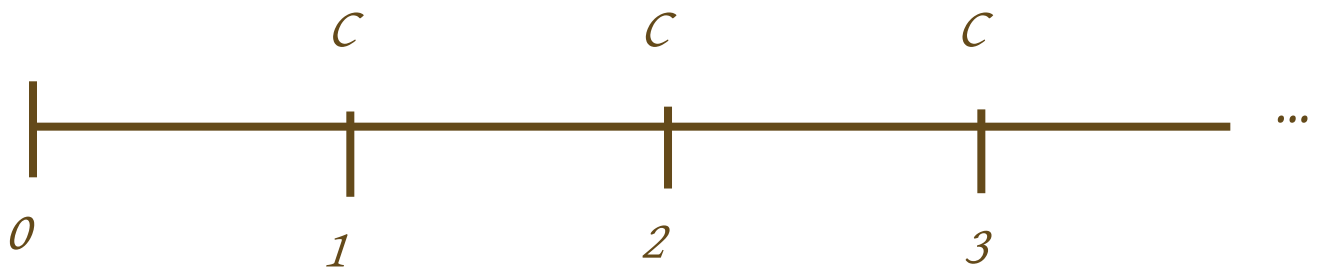
Perpetuity

- A constant stream of cash flows that lasts forever



$$PV = \frac{C}{(1+r)} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \cdots$$

- The formula for the present value of a perpetuity is:

$$PV = \frac{C}{r}$$

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Perpetuity - Example

- What is the value of a British Consol that promises to pay £15 each year, every year forever? The interest rate is 10%.

$$PV = \frac{£15}{0.10} = £150$$

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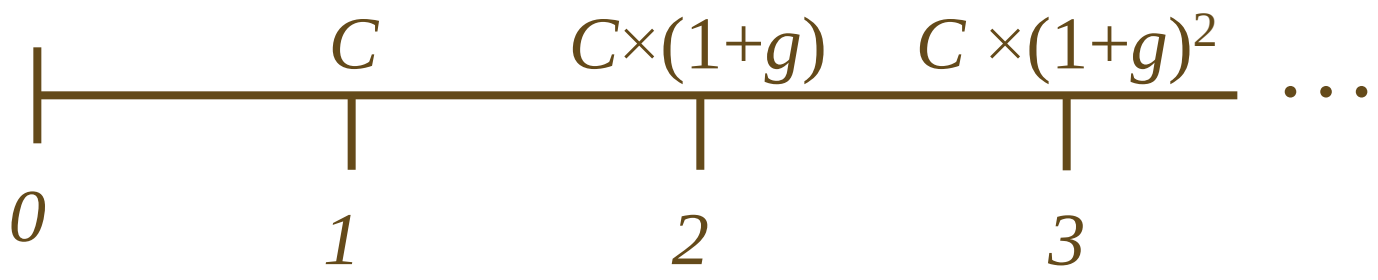
Growing Perpetuity

- A growing stream of cash flows that lasts forever



- The formula for the present value of a growing perpetuity is (if g<r):

$$PV = \frac{C}{(1+r)} + \frac{C \times (1+g)}{(1+r)^2} + \frac{C \times (1+g)^2}{(1+r)^3} + \cdots$$

$$PV = \frac{C}{r-g}$$

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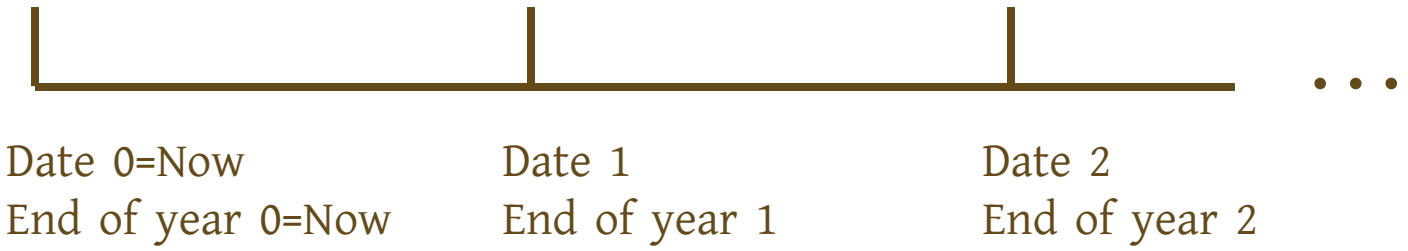
Growing Perpetuity

- Remarks on growing perpetuity formula:
  - C is cash flow at t=1
  - Require r > g
  - Assume a regular and discrete pattern of cash flow
- Inter-changeability of two conventions about time



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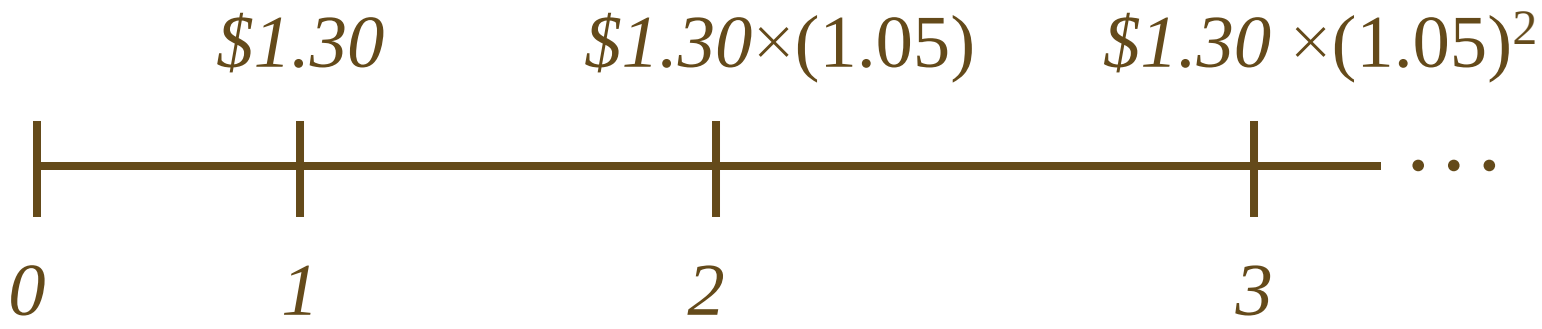
Growing Perpetuity - Example

- The expected dividend of Company XYZ next year is $1.30 and dividends are expected to grow at 5% forever. If the discount rate is 10%, what is the present value of this dividend stream?



$$PV = \frac{\$1.30}{0.10 - 0.05} = \$26.00$$

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PV of a Delayed Growing Perpetuity

- Your firm is about to make its initial public offering of stock and your job is to estimate the correct offering price. Forecast dividends are as follows:

Year:	1	2	3	4
Dividends per share	$1.50	$1.65	$1.82	5% growth thereafter

- If investors demand a 10% return on investments of this risk level, what price will they be willing to pay?

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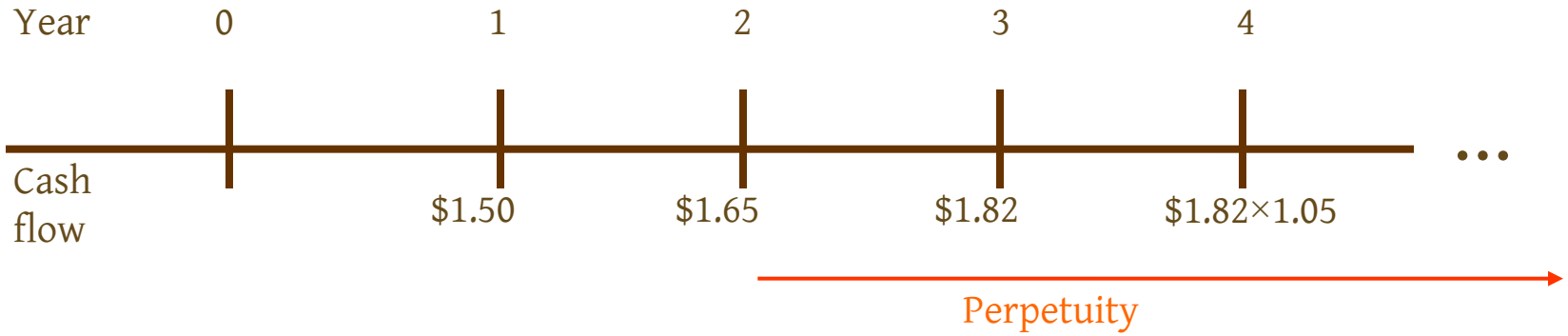
PV of a Delayed Growing Perpetuity



$$P_2 = \frac{C}{r-g} = \frac{1.82}{.10-.05} = \$36.40$$

$$P_0 = \frac{\$1.50}{(1.10)} + \frac{\$1.65+\$36.40}{(1.10)^2} = \$32.81$$

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Annuity

- A constant stream of cash flows with a fixed maturity

$$PV = \frac{C}{(1+r)} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \cdots \frac{C}{(1+r)^T}$$

- The formula for the present value of an annuity is: $$PV = \frac{C}{r}\left[1 - \frac{1}{(1+r)^T}\right]$$

- Annuity factor: $$A_r^T = \left[\frac{1}{r} - \frac{1}{r(1+r)^T}\right]$$

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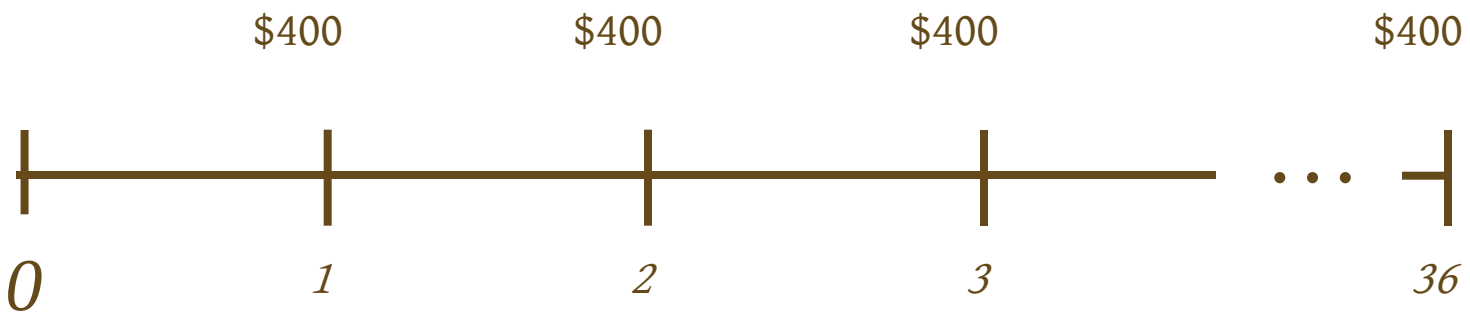
Annuity - Example

- If you can afford a $400 monthly car payment, how expensive a car can you buy if interest rates are 7% compounded monthly on 36-month loans?



$$PV = \frac{\$400}{0.07/12}\left[1 - \frac{1}{(1+0.07/12)^{36}}\right] = \$12{,}954.59$$

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Annuity: Canadian Mortgages

- What is special about Canadian Mortgages?
  - Canadian banks quote the annual interest compounded semi-annually for mortgages, although interest is calculated (compounded) every month.
  - The terms of the mortgage are usually renegotiated before the end of the mortgage. For example, the interest of a 25-year mortgage can be renegotiated 5 years after the initiation of the mortgage.

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Canadian Mortgages - Example

- You have negotiated a 25-year, $100,000 mortgage at a rate of 7.4% per year compounded semi-annually with the Toronto-Dominion Bank. What is your monthly payment?
- To answer this question, we first have to convert the quoted mortgage rate to the effective annual rate:

$$r_e = \left(1 + \frac{r_s}{m}\right)^m - 1 = \left(1 + \frac{0.074}{2}\right)^2 - 1 = 7.5369\%$$

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Canadian Mortgages - Example

- The effective interest rate charged each month is given by:

$$(1+r_e)^{1/m}-1=\left(1+0.075369\right)^{\frac{1}{12}}-1=.607369\%$$

- We use the present value of annuity formula to find the monthly payment C:

$$PV=\frac{C}{r}\left[1-\frac{1}{(1+r)^T}\right]$$

$$100,000=\frac{C}{0.00607369}\left[1-\frac{1}{(1+0.00607369)^{300}}\right]$$

$$C=\$725.28$$

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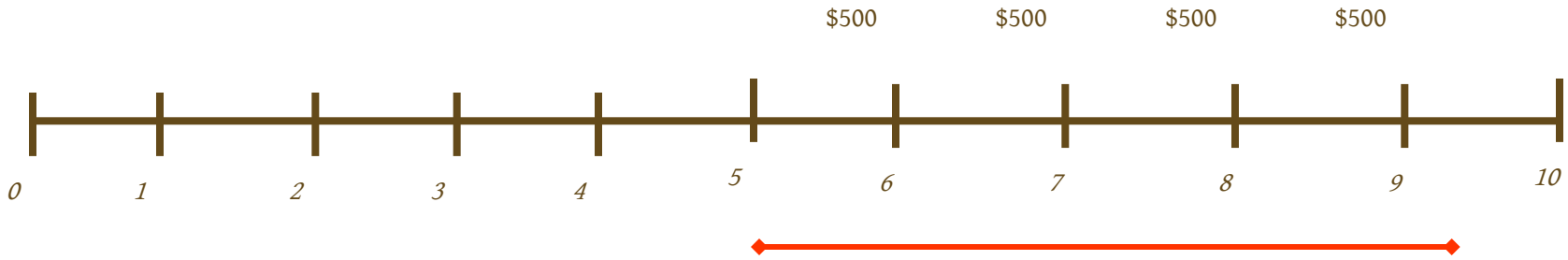
Delayed Annuity

- You receive a four-year annuity of $500 per year beginning at date 6. r=10%. What's PV of the annuity?



- Calculate PV of annuity in year 5

$$\$500\left[\frac{1}{0.1}-\frac{1}{0.1(1+0.1)^4}\right]=\$500\times A_{0.1}^{4}=\$1,584.95$$

- Discount the PV back to year 0

$$\frac{\$1,584.95}{(1+0.1)^5}=\$984.13$$

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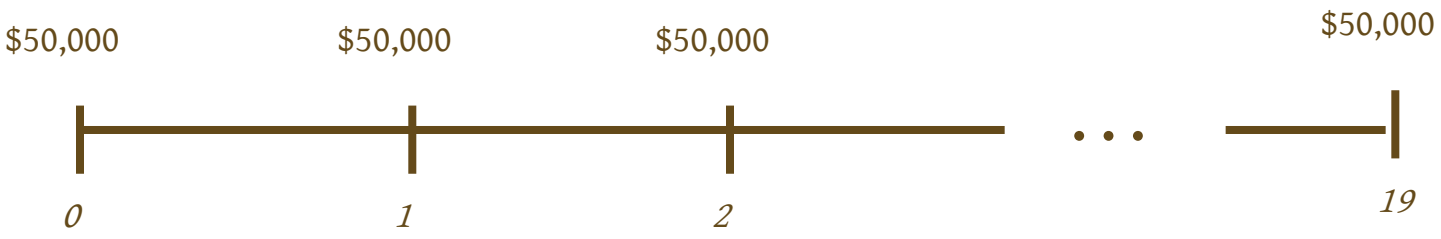
Annuity in Advance

- You'll receive $50,000 a year for 20 years. The first payment occurs immediately. r=8%



PV = Payment in year 0 + 19-year annuity

$$= \$50{,}000 + \$50{,}000 \times A^{19}_{0.08}$$

$$= \$530{,}180$$

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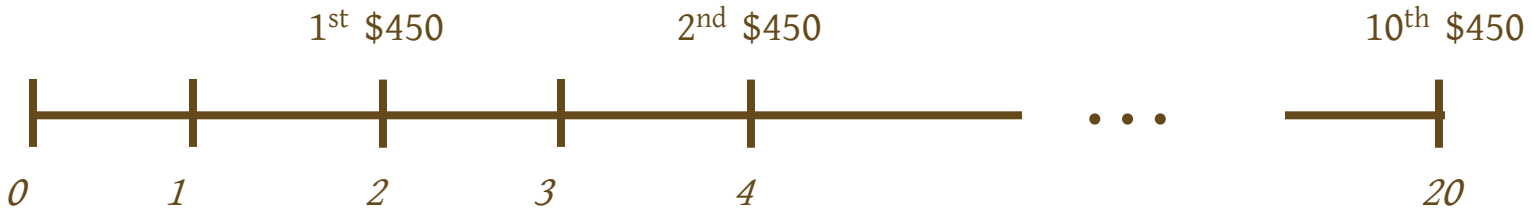
Infrequent Annuity

- You receives an annuity of $450 payable once every two years for 20 years. r=6%. What's the present value of the annuity?



- Determine the interest rate over a 2-year period: (1.06×1.06)–1 = 12.36%

- PV of a $450 annuity over 10 periods:

$$\$450 \times A^{10}_{0.1236} = \$2{,}505.57$$

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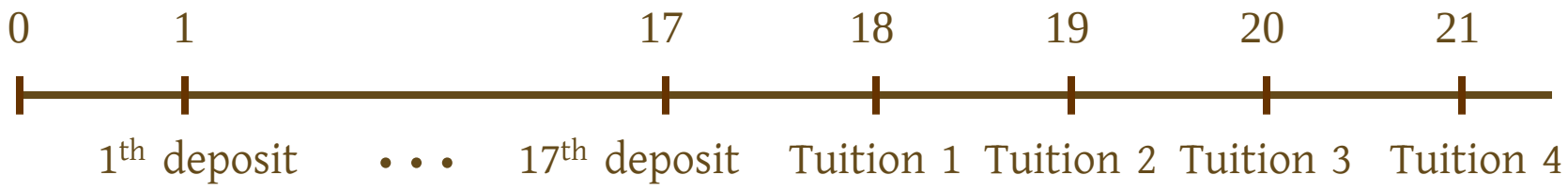
Equating PV of Two Annuities

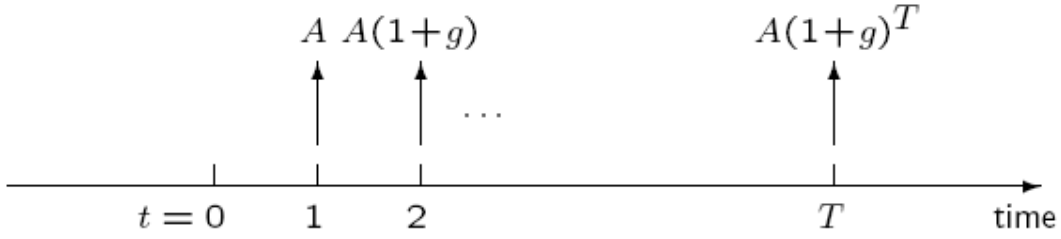
- Parents are saving for the university of their newborn daughter. Estimated expenses will be \$30,000 per year when the daughter is 18. r=14%. How much should the parents deposit to support a 4-year university education?



- PV of 4-year at university in year 17: \$30,000 $\times$ $A^{4}_{0.14}$ = \$87,411
- PV of a university education in year 0: $\dfrac{\$87,411}{(1+0.14)^{17}} = \$9,422.91$
- Annual deposit that will yield PV of \$9,422.91: C $\times$ $A^{17}_{0.14}$\$9,422.91 $\Rightarrow$ C=\$1,478.59

Growing Annuity

- A growing stream of cash-flow with a fixed maturity



PV (growing annuity)

$$= A \times \left[\frac{1}{1+r} + \frac{1+g}{(1+r)^2} + \cdots + \frac{(1+g)^{T-1}}{(1+r)^T} \right]$$

$$= A \times \begin{cases} \frac{1}{r-g}\left[1 - \left(\frac{1+g}{1+r}\right)^T\right] & \text{if } r \neq g \\ \frac{T}{1+r} & \text{if } r = g. \end{cases}$$

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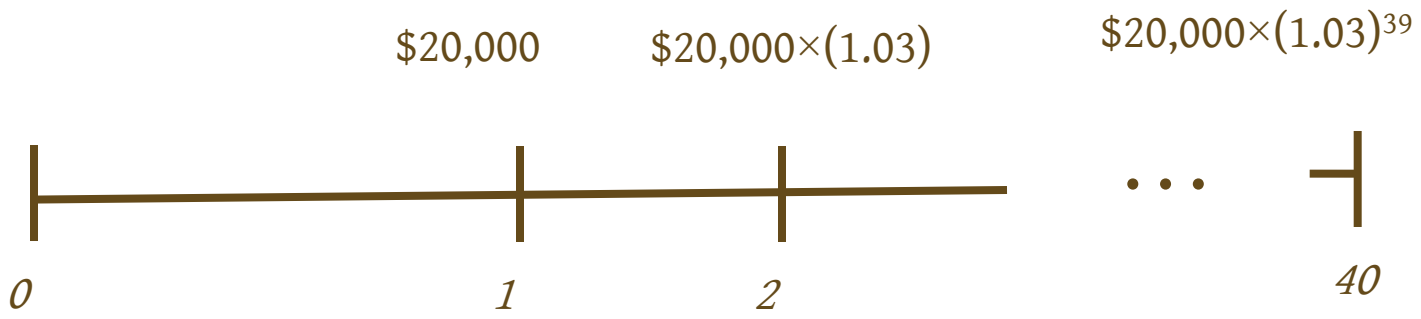
Growing Annuity - Example

- A retirement plan offers to pay \$20,000 the first year, and to increase the annual payment by 3% each year until you die. Assume you plan to die in 40 years. What is the present value at retirement if the discount rate is 10%?



$$PV = \frac{\$20{,}000}{0.10 - 0.03}\left[1 - \left(\frac{1.03}{1.10}\right)^{40}\right] = \$265{,}121.57$$

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Summary

- Future value, present value, r_s and r_e
- The net present value formula:

$$NPV = -C_0 + \frac{C}{(1+r)} + \frac{C}{(1+r)^2} + \cdots + \frac{C}{(1+r)^T} = -C_0 + \sum_{t=1}^{T} \frac{C}{(1+r)^t}$$

- Four simplifying formulas:

Perpetuity: $PV = \frac{C}{r}$

Growing Perpetuity: $PV = \frac{C}{r-g}$

Annuity: $PV = \frac{C}{r}\left[1 - \frac{1}{(1+r)^T}\right]$

Growing Annuity: $PV = \frac{C}{r-g}\left[1 - \left(\frac{1+g}{(1+r)}\right)^T\right]$

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Stock Valuation

Sobey School of Business
Saint Mary's University

Basic Model 1: Zero Growth DDM

- Assume that dividends will remain at the same level forever:

$$Div_1 = Div_2 = Div_3 = \cdots = Div_\infty$$

- Since future cash flows are constant, the value of a zero-growth stock is the present value of a perpetuity:

$$\Rightarrow P_0 = \frac{Div}{(1+r)^1} + \frac{Div}{(1+r)^2} + \frac{Div}{(1+r)^3} + \cdots$$

$$\Rightarrow P_0 = \frac{Div}{r}$$

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Basic Model 2: Constant Growth DDM

- Assume that dividends will grow at a constant rate, g, forever. The dividend flow looks like the following:

$$
\begin{aligned}
&D_1 = D_0 \times (1+g) \\
&D_2 = D_1 \times (1+g) = D_0 \times (1+g) \times (1+g) = D_0 \times (1+g)^2 \\
&D_3 = D_2 \times (1+g) = D_0 \times (1+g)^2 \times (1+g) = D_0 \times (1+g)^3 \\
&. \\
&D_T = D_{T-1} \times (1+g) = D_0 \times (1+g)^{T-1} \times (1+g) \\
&= D_0 \times (1+g)^T
\end{aligned}
$$

- Since future cash flows grow at a constant rate forever, the value of a constant growth stock is the present value of a growing perpetuity:

$$P_0 = \frac{D_1}{r-g} = \frac{D_0(1+g)}{r-g}$$

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Basic Model 3: Differential Growth DDM

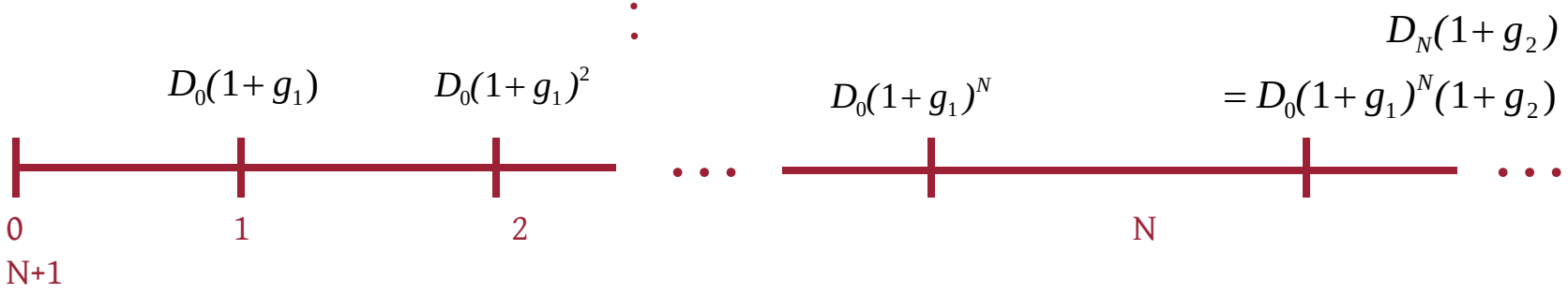
- Assume that dividends will grow at rate g_1 for N years and grow at rate g_2 thereafter:

$$D_1 = D_0 \times (1+g_1)$$
$$D_2 = D_1 \times (1+g_1) = D_0 \times (1+g_1)^2$$

$$\vdots$$

$$D_N = D_{N-1} \times (1+g_1) = D_0 \times (1+g_1)^N$$
$$D_{N+1} = D_N \times (1+g_2) = D_0 \times (1+g_1)^N \times (1+g_2)$$

$$\vdots$$



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Basic Model 3: Differential Growth DDM

- We can value this as the sum of an N-year annuity growing at rate g_1...

$$P_A = \frac{D_1}{r - g_1} \times \left[1 - \frac{(1+g_1)^N}{(1+r)^N}\right]$$

...plus the discounted value of a perpetuity growing at rate g_2 that starts in year N

$$P_B = \frac{\left(\frac{D_{N+1}}{r - g_2}\right)}{(1+r)^N} = \frac{D_1 \times (1+g_1)^{N-1} \times (1+g_2)}{(r - g_2) \times (1+r)^N}$$

$$P_0 = P_A + P_B$$

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Basic Model 4: Geometric Stochastic DDM

- The geometric model assumes that dividends increase at a constant growth rate or stay unchanged. Letting g be the growth rate. Assume
  - $D_{t+1} = D_t(1 + g)$ with probability p
  - $D_{t+1} = D_t$ with probability 1 – p
- The intrinsic value of the stock is:

$$P_0 = \frac{D_0(1 + pg)}{r - pg}$$

  - Note that *pg* is the expected growth rate.

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Basic Model 5: Additive Stochastic DDM

- The additive model assumes that dividends increase by a constant amount or stay unchanged.
  - $D_{t+1} = D_t + C$ with probability p
  - $D_{t+1} = D_t$ with probability 1 – p

- The theoretical value of the stock is:

$$P_0 = \frac{D_0}{r} + \left(\frac{1}{r} + \frac{1}{r^2}\right) Cp$$

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Basic Model 6: Allowing For Dividend Cut

- It is not uncommon for firms to cut dividends temporarily.

- The geometric model can be revised as
  - $D_{t+1} = D_t(1 + g)$ with probability p_u
  - $D_{t+1} = D_t(1 - g)$ with probability p_d
  - $D_{t+1} = D_t$ with probability $1 - p_u - p_d$

- The theoretical value of the stock is:

$$P_0 = \frac{D_0[1 + (p_u - p_d)g]}{r - (p_u - p_d)g}$$

  - Note that $(p_u - p_d)g$ is the expected growth rate.

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Basic Model 7: Allowing For Dividend Cut

- The additive model can be revised as
  - $D_{t+1} = D_t + C$ with probability p_u
  - $D_{t+1} = D_t - C$ with probability p_d
  - $D_{t+1} = D_t$ with probability $1 - p_u - p_d$

- The theoretical value of the stock is:

$$P_0 = \frac{D_0}{r} + \left(\frac{1}{r} + \frac{1}{r^2}\right) C(p_u - p_d)$$

- There is no evidence that the geometric model for dividends is superior or inferior to the additive model.

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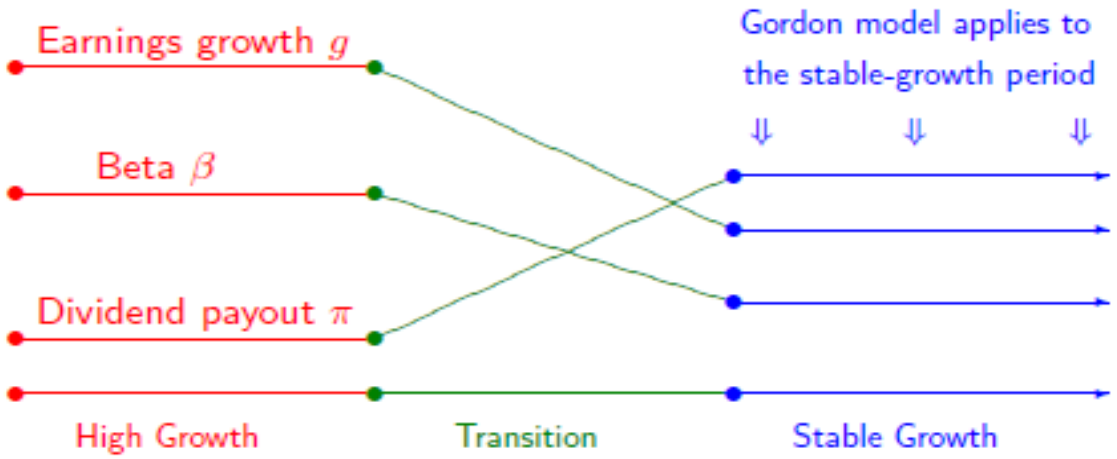
Basic Model 8: Multi-Stage DDM

- A popular extension of the Gordon model is the three-stage DDM.
- The three-stage DDM allows for an initial period of high growth, a transitional period when growth declines, and a final stable-growth phase.



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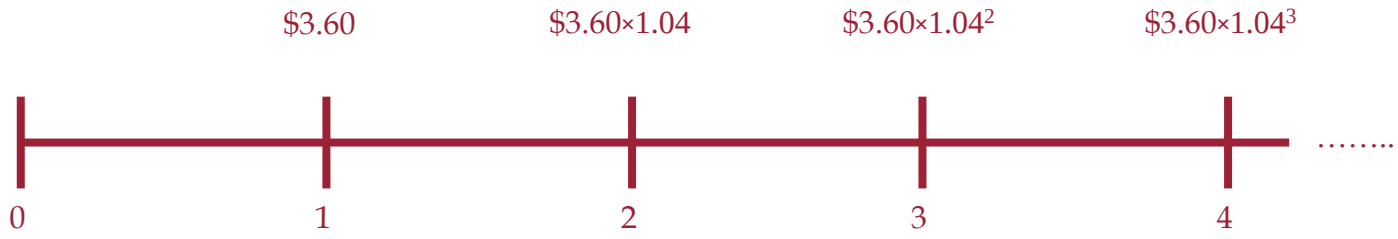
Traditional Valuation Model -- Example

- XYZ Corp. has a common stock that paid its annual dividend of **$3.50** this morning. It is expected to pay a **$3.60** dividend one year from now, and following dividends are expected to grow at a rate of **4%** per year forever. If stocks of similar risk earn **16%** per year, what is the price of a share of XYZ stock?
- The stock price is given by the present value of the perpetual stream of growing dividends:



$$P_0 = \frac{D_1}{r-g} = \frac{D_0(1+g)}{r-g} = \frac{3.60}{0.16-0.04} = \$30.00$$

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Beyond Basics: Question?

- How confident are you that XYX Inc. stock will indeed be $30 one year from now?

- In other words, what is the probability that XYX Inc. stock will indeed be $30 one year from now?

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A Traditional NPV Model

- Initial investment outlay = $12,000
- After-tax cash incremental flow benefits:
  - Year 1 = $5,000
  - Year 2 = $5,000
  - Year 3 = $8,000
- Discount rate (r) = 15%

$$
\begin{aligned}
NPV &= \frac{CF_1}{(1+r)^1} + \frac{CF_2}{(1+r)^2} + \frac{CF_3}{(1+r)^3} - CF_0 \\
&= \frac{\$5{,}000}{(1.15)^1} + \frac{\$5{,}000}{(1.15)^2} + \frac{\$8{,}000}{(1.15)^3} - \$12{,}000 \\
&= \$4{,}348 + \$3{,}781 + \$6{,}260 - \$12{,}000 \\
&= \$1{,}389
\end{aligned}
$$

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Beyond Basics: Question?

- How confident are you that NPV will indeed be $1389 from this project?
- In other words, what is the probability that NPV of the project will indeed be $1389?

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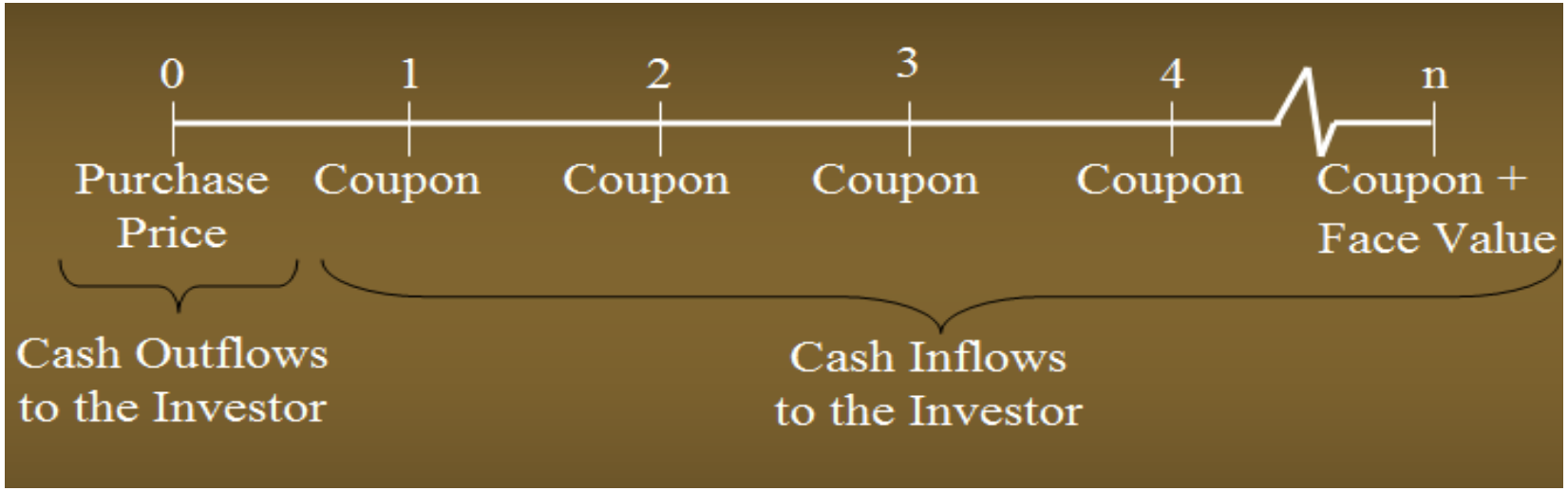
Bond Valuation

Sobey School of Business
Saint Mary's University

Basic Structure: Cash Flow Pattern



- The Purchase Price or Market Price of a bond is simply the present value of the cash inflows, discounted at the bond's yield-to-maturity

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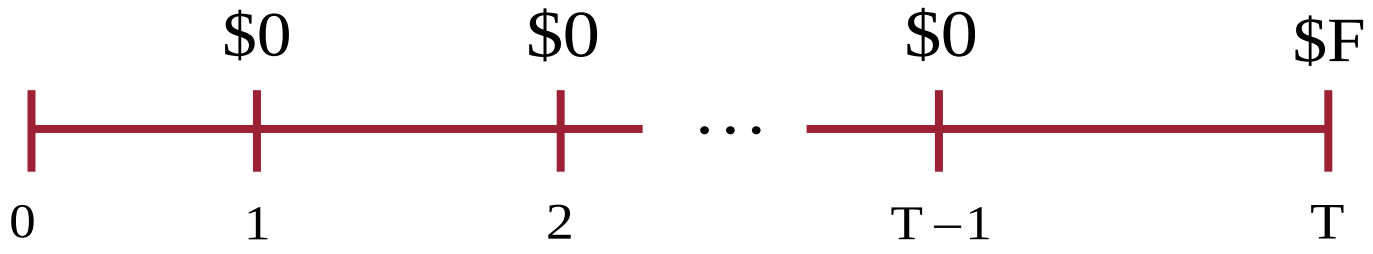
Bond Valuation: Pure Discount Bonds

- Definition: pure discount (or zero-coupon) bonds do not pay coupons
- Information needed for valuing pure discount bonds:
  - Time to maturity (T) = Maturity date - today's date
  - Face value (F)
  - Discount rate (r)



- Present value of a pure discount bond at time 0:

$$PV = \frac{F}{(1+r)^T}$$

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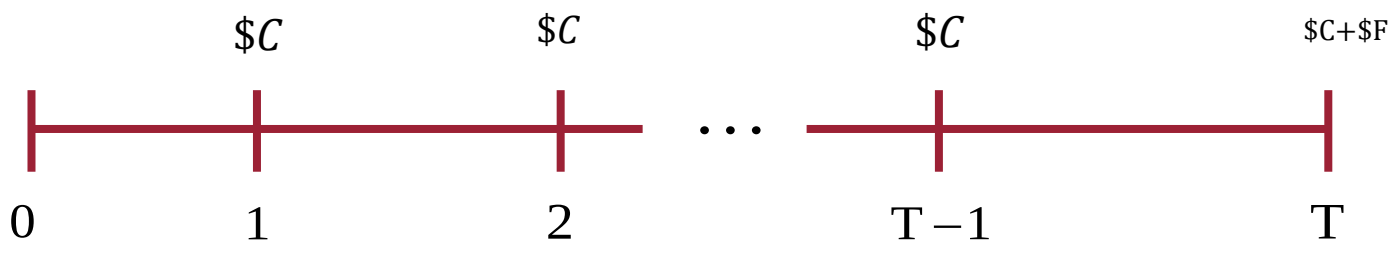
Bond Valuation: Coupon Bonds

- Definition: A coupon bond pays coupon payment during the term of the bond.
- Information needed for valuing pure discount bonds:
  - Time to maturity (T) = Maturity date - today's date
  - Face value (F)
  - Discount rate (r)
  - Couple payment (C)



Present value of a pure discount bond at time 0:

$$\mathrm{PV} = \left[\frac{C}{r} \times \left(1 - \frac{1}{(1+r)^T}\right)\right] + \frac{\mathrm{F}}{(1+\mathrm{r})^{\mathrm{T}}}$$

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Bond Property - I

- Bond prices and market interest rates move in opposite directions.
- Trading strategy:
  - If you expect the interest rate to go up in the future, sell.
  - If you expect the interest rate to go down in the future, buy

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Bond Property - II

- Bond prices and market interest rates move in opposite directions
  - When coupon rate = r, price = par value.
  - When coupon rate > r, price > par value (premium bond)
  - When coupon rate < r, price < par value (discount bond)

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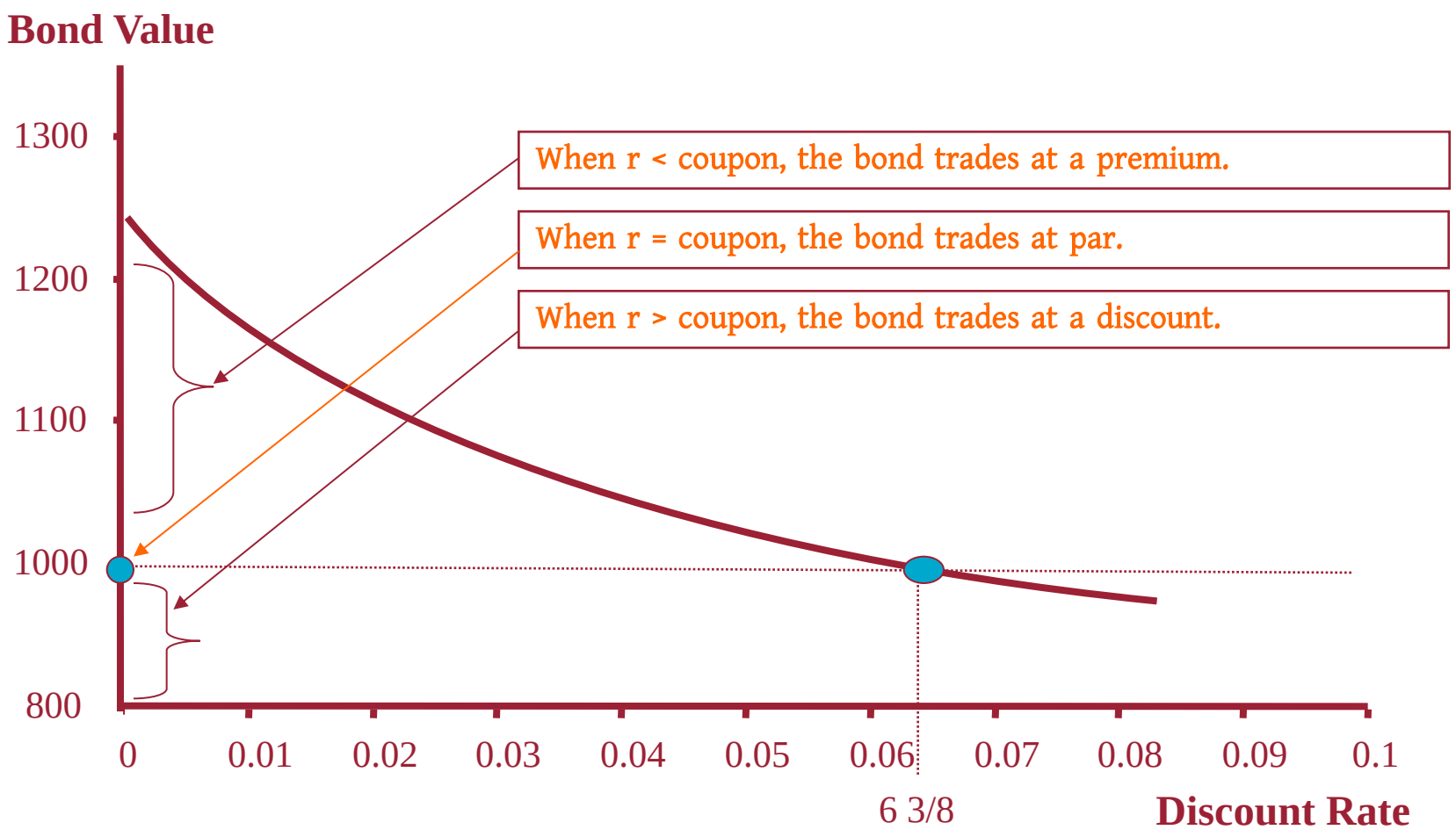
Discount Rate and Bond Value



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YTM: Example

- Consider a bond with F=$1000, C=$100 (paid annually), and T=30
  - Suppose that the current price is $1,000
    - What is the YTM?
    - This is easy – we know it has to be 10%
  - Suppose instead that the price is $1001.68
    - The bond sells at premium, so we know y<10%
    - Solve for YTM either by trial and error, a calculator, or Excel (use the IRR function)
    - y = 9.9% < 10%

- With a zero coupon bond it is easier to solve for y
  - $P_{t,T} = F/(1+y)^T$
  - Minor adjustment if compounding is semi-annual, monthly, continuous, etc.

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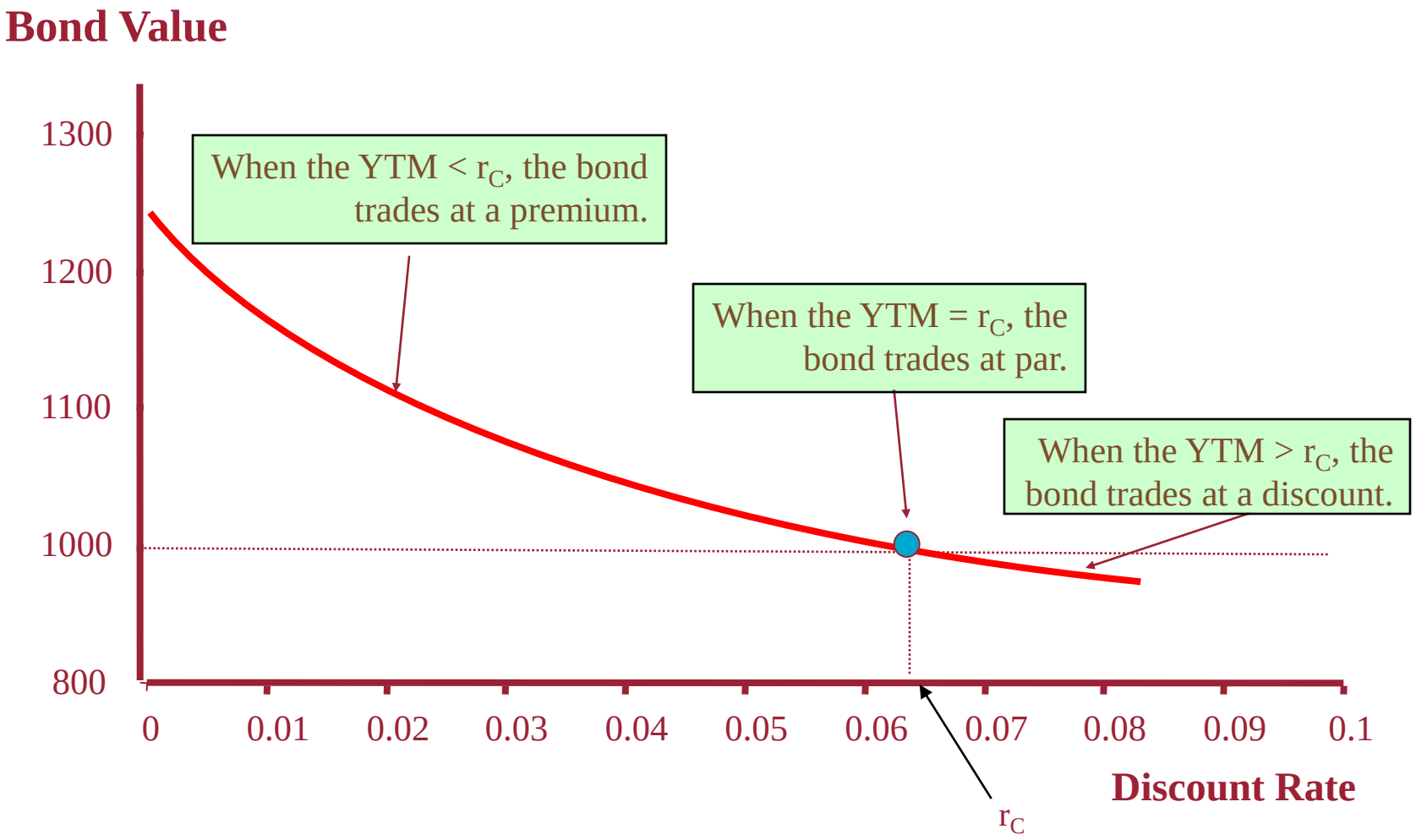
YTM and Bond Value



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Bond Price and Interest Rate Risk

- The risk that the bond price, $P_{t,T}$, will change due to changes in the interest rate.

- Two dimensions of the interest rate risk:
  - Time to maturity, T;
    - Intuition: As T increases (decreases), the compounding effect will be greater (smaller), leading to larger (smaller) changes in $P_{t,T}$ as r changes.
  - Coupon rate, r_C.
    - Intuition: As r_C increases (decreases), less (more) of the bond's value comes from the face amount, leading to smaller (larger) changes in $P_{t,T}$ as r changes.

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Interest Rate Risk: T

- Example: consider 2 bonds with r_C = 10%, but T_1 = 3 and T_2 = 20. Assume initially that the interest rate is such that both bond prices are equal to 1,000.
  - What is the initial interest (discount) rate?
  - What are the bond prices if interest rate increases to 11%?
  - What if interest rate decreases to 9%?
    - For r=11%, $P_{0,3}$ = \$975.56, $P_{0,20}$ = \$920.37;
      (%Δ = -2.4%) (%Δ = -8%)
    - For r=0.09, $P_{0,3}$ = \$1,025.31, $P_{0,20}$ = \$1,091.29
      (%Δ = +2.5%) (%Δ = +9.1%)
  - Which bond is more risky? (more volatile)

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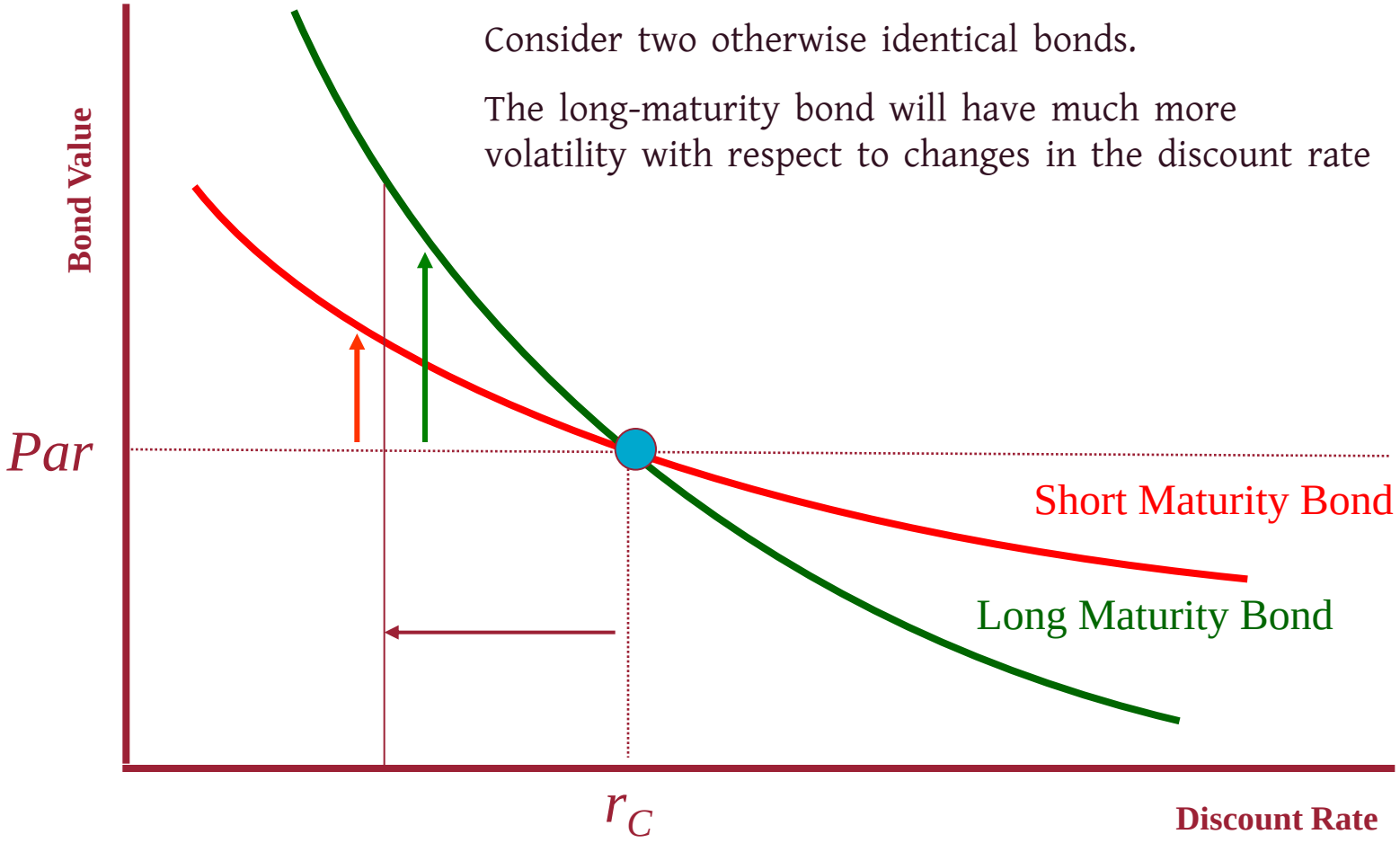
Maturity and Bond Price Volatility

Consider two otherwise identical bonds.

The long-maturity bond will have much more volatility with respect to changes in the discount rate



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Bond Property - III

- Bond prices and market interest rates move in opposite directions.
- A bond with longer maturity has higher relative (%) price change than one with shorter maturity when interest rate changes.

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Interest Rate Risk: r_C

- Example: consider 2 bonds with T = 10 but r_C^1 = 10%, r_C^2 = 3%. Assume initially that the interest rate is 10%.

  - What are the initial bond prices?
  - $P1_{0,10}(r_C=10\%)$ = \$1,000, $P2_{0,10}(r_C=3\%)$ = \$569.88;

  - What are the bond prices if interest rate increases to 11%? What if interest rate decreases to 9%?

  - For r=11%, $P1_{0,10}$ = \$941.11, $P2_{0,10}$ = \$528.86;
  - (%Δ = -5.9%) (%Δ = -7.2%)
  - For r=0.09, $P1_{0,10}$ = \$1,064.18, $P2_{0,10}$ = \$614.94
  - (%Δ = +6.4%) (%Δ = +9.1%)

  - Which bond is more risky? (more volatile)

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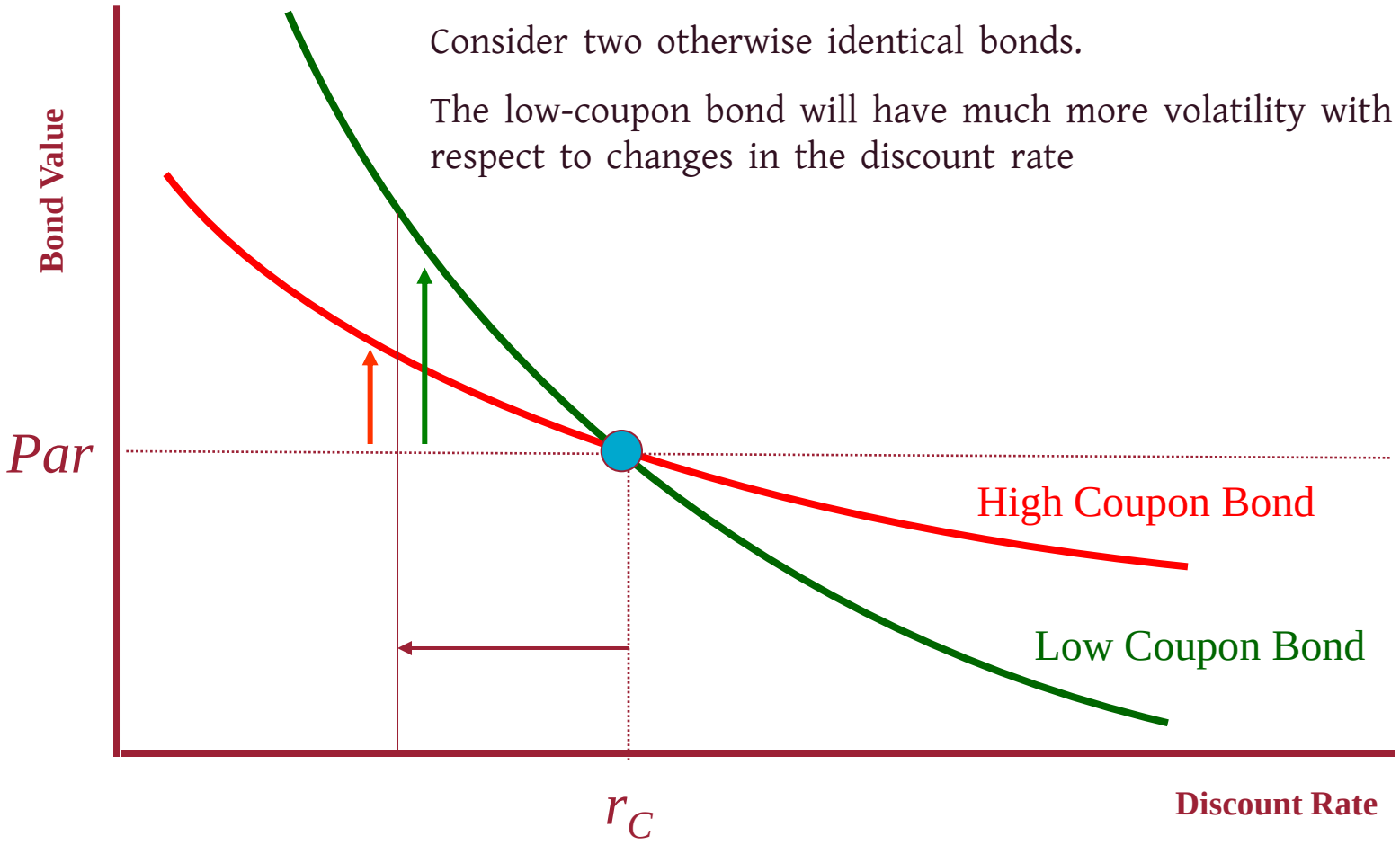
Coupon Rate and Bond Price Volatility

Consider two otherwise identical bonds.

The low-coupon bond will have much more volatility with respect to changes in the discount rate



Bond Property - IV

- Bond prices and market interest rates move in opposite directions.
- A lower coupon bond has a higher relative price change than a higher coupon bond when interest rate changes.

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Holding Period Return (HPR)

- Suppose that on January 1, 2002, you bought a 6.375 coupon Government of Canada bond with semi-annual payments, and a maturity date of December 31, 2009.
- At that time the YTM was 5-percent, and you paid \$1,089.75 ($P_{0,8}$).
- Six months later (July 1, 2002), You sold the bond when the YTM was 4-percent.
- What is the holding period return?

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Bond Property - V

- The holding period return for a coupon bond equals its yield to maturity only if you can reinvest the coupons at the yield to maturity.
  - Reinvestment at a higher rate produces a holding period return (realized coupon yield) above YTM,
  - Reinvestment at a lower rate produces a holding period return below YTM.

- For a zero coupon bond the yield to maturity is always equal to the holding period return if the bond is held to maturity.

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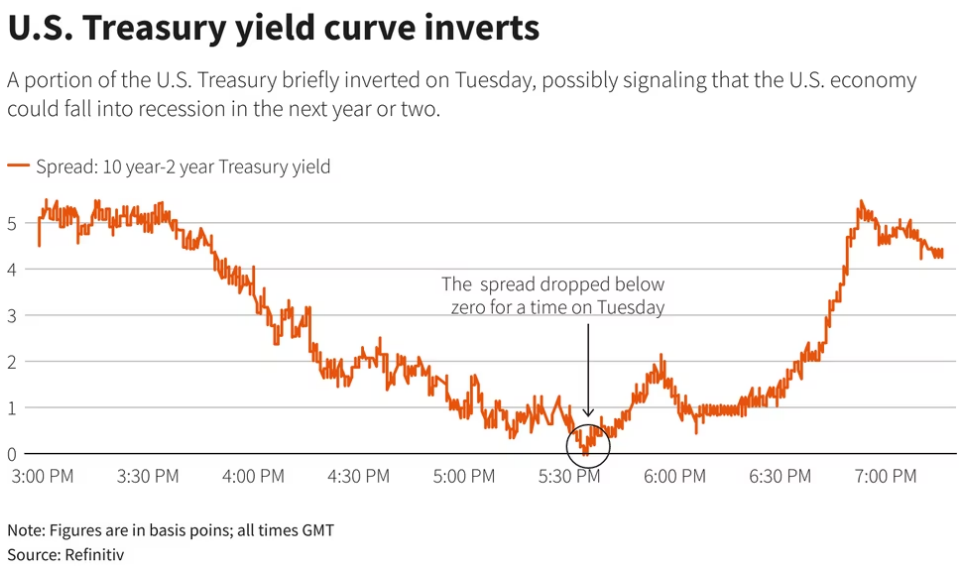
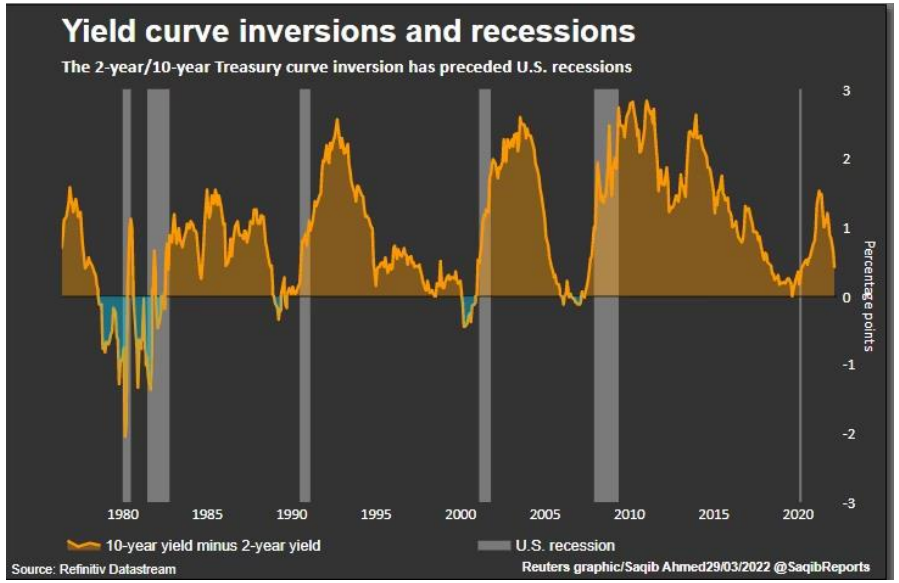
The Term-structure of Interest Rates

- The **term structure of interest rates** is the relationship between 'time to maturity' (T) and 'yield to maturity' (YTM):
- Term structure gives you the schedule of spot rates for each maturity. It is also known as **Yield Curve**.



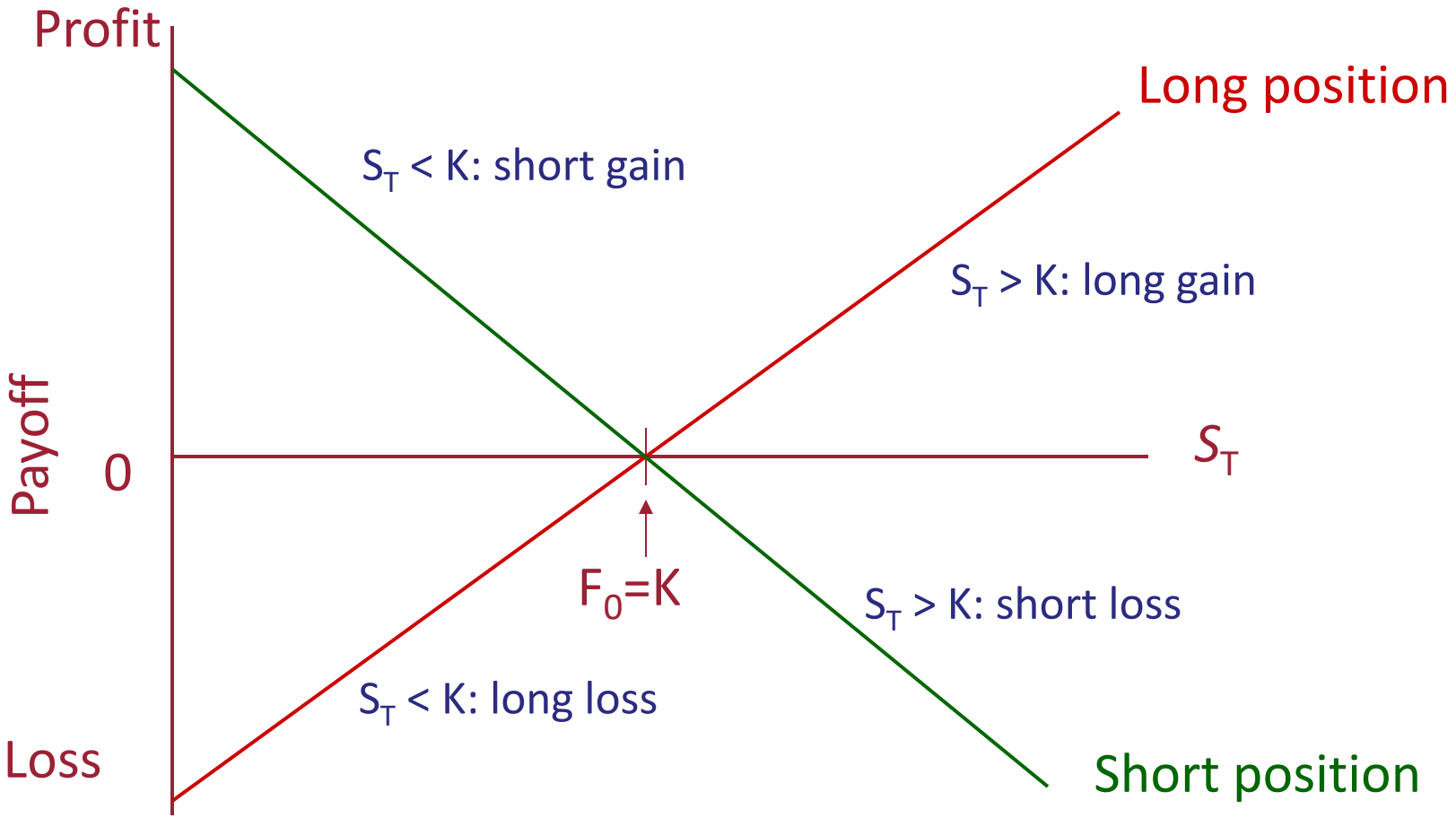


Financial Derivatives

Sobey School of Business
Saint Mary's University

Forward Contract Payoffs

Zero-sum game: the long position payoff is the opposite of the short



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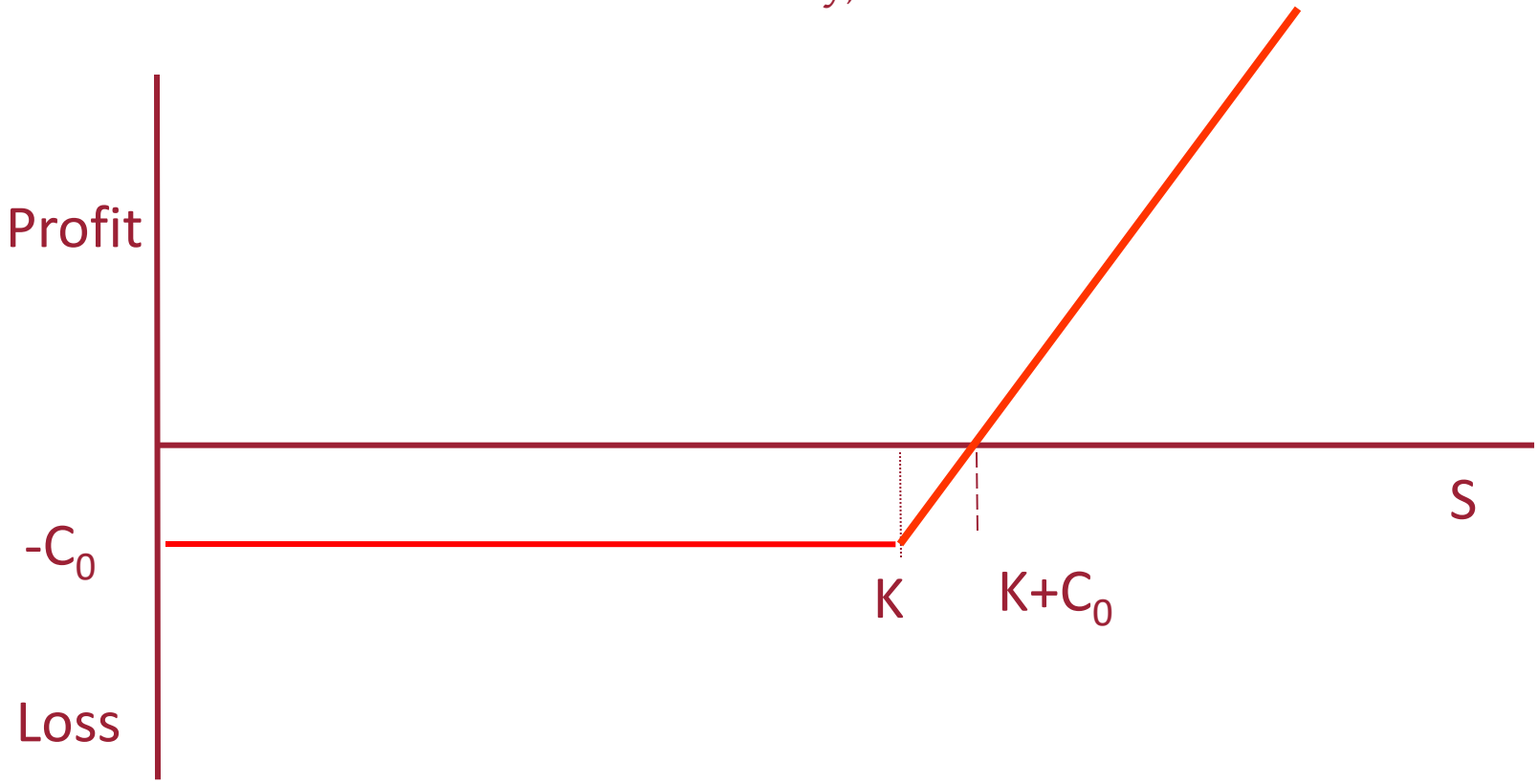
Profit: Long 1 Call Option

At expiration, $C_T = \text{Max}[S_T - K, 0]$

If the call is in-the-money, it is worth $S_T - K$ (payoff)

Option buyer's profit = Payoff – C_0

If the call is out-of-the-money, it is worthless



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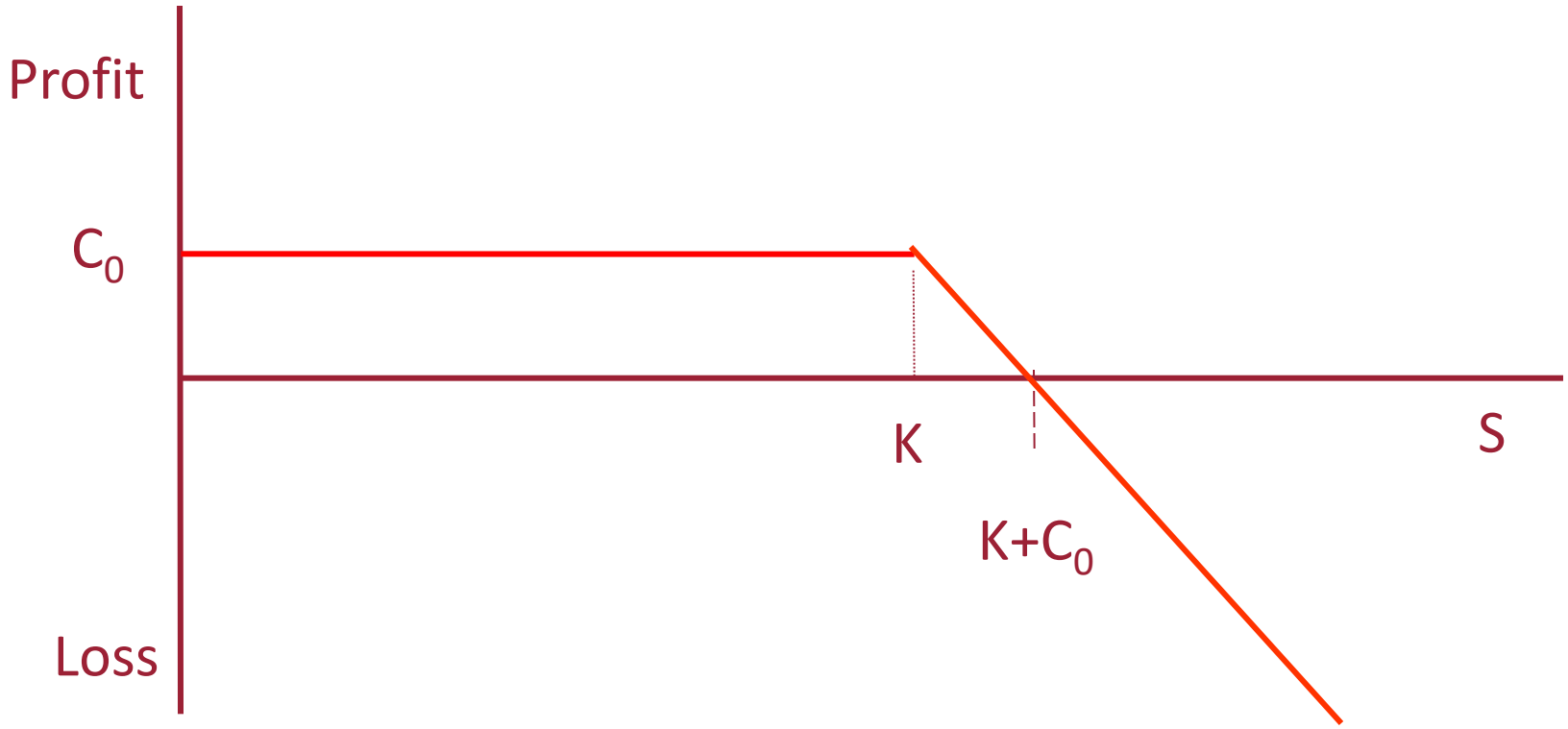
Profit: Short 1 Call Option

At expiration, $C_T = \text{Max}[S_T - K, 0]$

Option writer (short) is paid C_0

Option writer's profit: $C_0 - (S_T - K)$

Makes C_0 if call is out of the money



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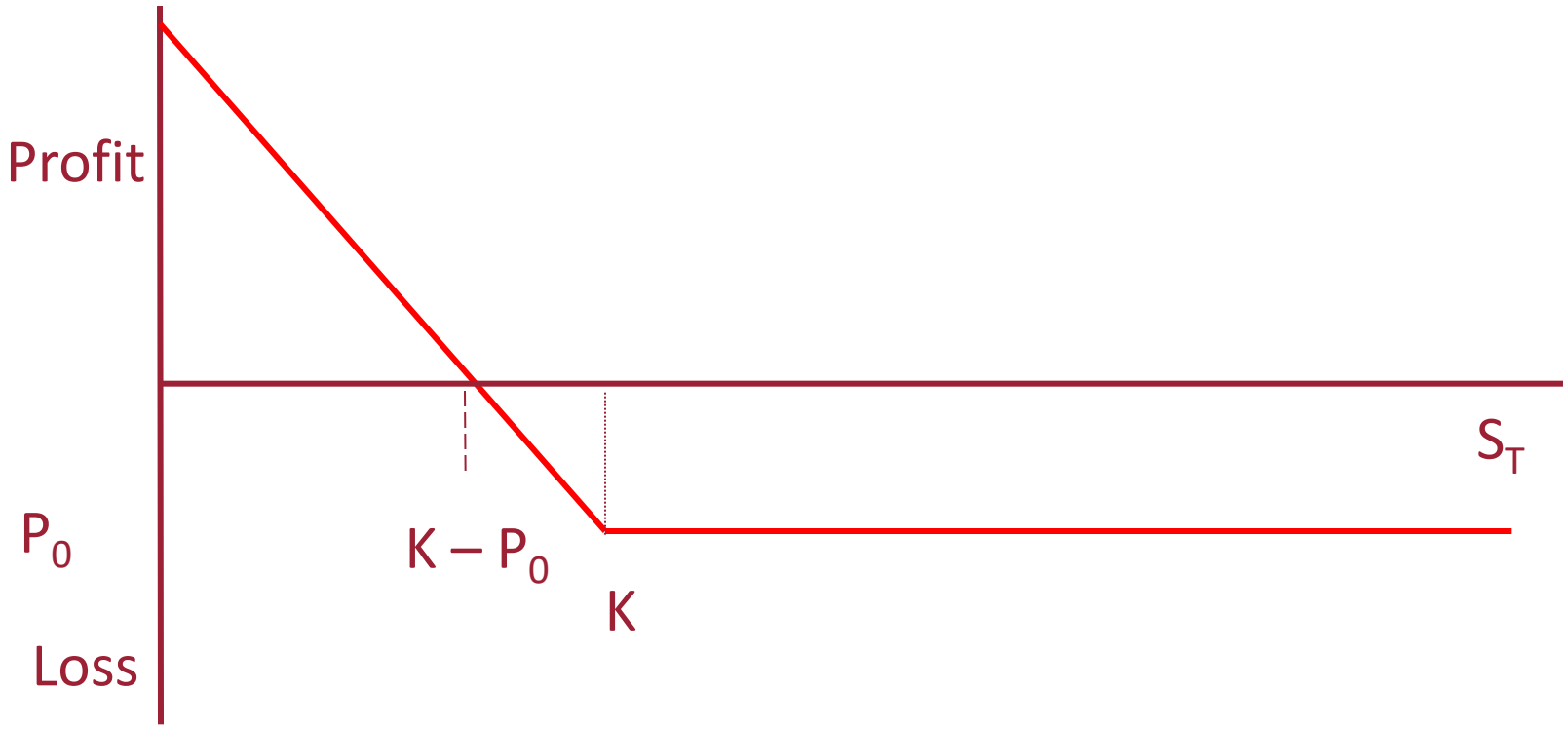
Profit: Long 1 Put Option

At expiration, $P_T = \text{Max}[K - S_T, 0]$



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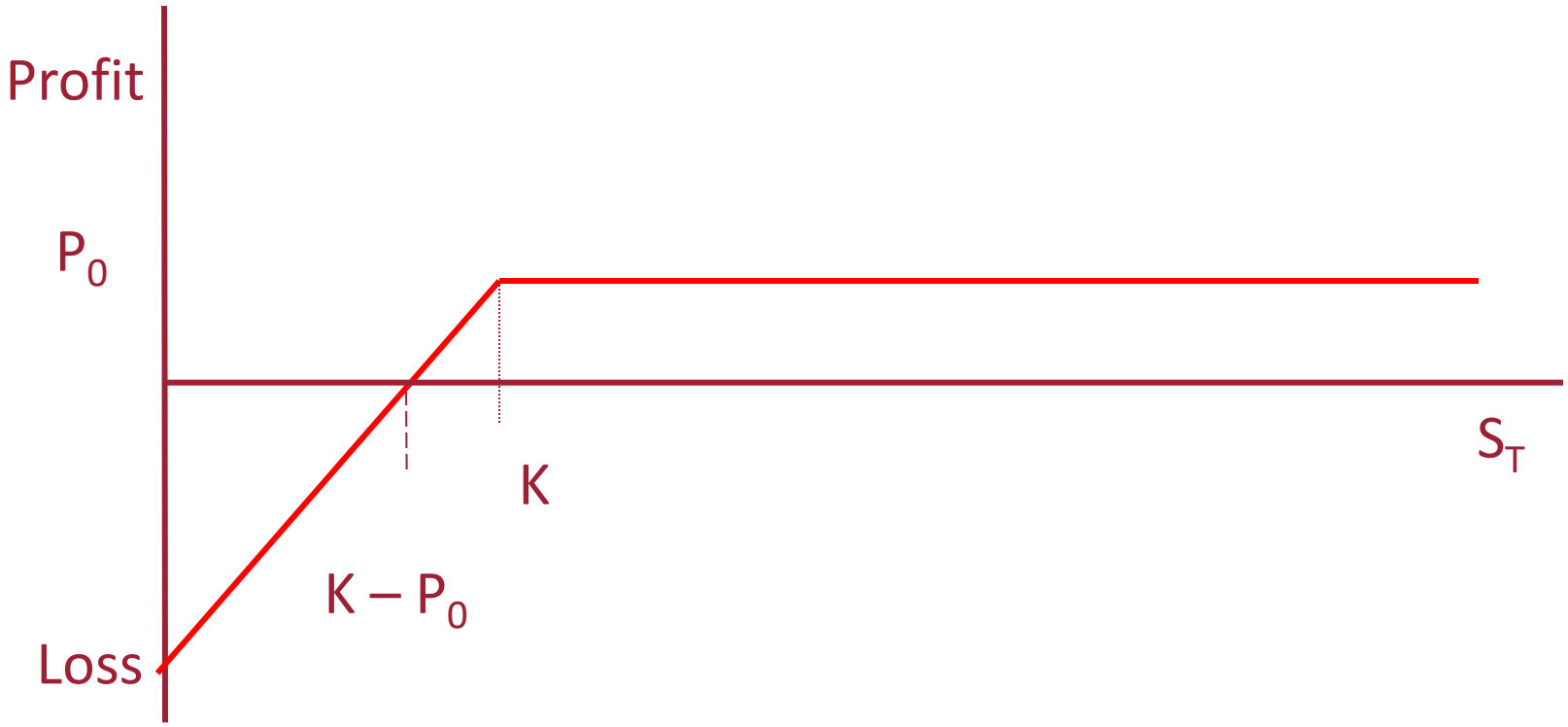
Profit: Short 1 Put Option

At expiration, $P_T = \text{Max}[K - S_T, 0]$



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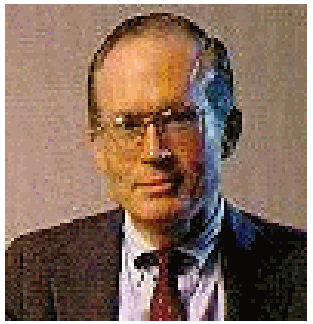
Option Pricing

- Option holders do not have to exercise their option
  - Distinguishes options from forwards and futures
  - This gives options value, so there is a cost to enter a long position in an option contract – called a premium
    - **Intrinsic Value:** the difference between the exercise price of the option and the spot price of the underlying asset (for calls, $S_t - K$)
    - **Time Value:** the difference between the option premium and the intrinsic value of the option

Option Premium	=	Intrinsic Value	+	Time Value



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